Class 6

Biology

(Solutions)

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1. Food - Where Does It Come From?

Solutions

LEVEL-I

1. Correct option (D)

A condiment is a food item that is added to another food item to enhance its flavor. Due to its widespread usage in enhancing food flavor, pepper is regarded as a condiment. Pepper is a flavorful spice that is used in many different cuisines to add heat, depth, and complexity to foods. The Piper nigrum plant's dried, crushed, or powdered peppercorns are used to make it.

Pepper is one of the most well-known and adaptable spices in the world and has been used as a condiment for ages. It gives a food a unique flavor and can be a key element in creating the appropriate flavor profile.

2. Correct option (D)

Ingredients, cooking technique, and regional cultural customs are just a few of the variables that affect how food tastes.

It goes without saying that a dish's flavor will be greatly influenced by the items utilized. For instance, a food cooked with soy sauce, ginger, and sesame oil will taste considerably different than a dish made with tomatoes, basil, and garlic.

The way food is prepared can also impact how it tastes. For instance, a rare steak will taste quite different from a well-done steak.

The regional cultural customs can have an impact on how food tastes. For instance, while adding spices to meals is prevalent in certain cultures, it is less frequent in others.

3. Correct option (D)

When purchasing packed food, it is important to consider all of the following:

a) Ingredients: Checking the list of ingredients is crucial to understand what the food product contains. This helps in determining if there are any specific ingredients you may want to avoid due to allergies, dietary restrictions, or personal preferences.

b) Manufacturing date: The production date tells us when the product was created. In particular for perishable commodities, it might be helpful for determining the freshness and shelf life of the product. The production date may be less important for non-perishable commodities, though.

c) Expire: The recommendation for when the product should be consumed is indicated by the expiration date. It assists in determining the food's safety and freshness. Foodborne infections might result from consuming food after it has expired, which is dangerous.

4. Correct option (B)

Almonds are not typically considered a fragrant material. Almonds do have a light nutty flavor, but unlike the other items on the list, they are not particularly well-known for their robust scent.

A) Cinnamon: This spice is well-known for its distinctive, pleasantly warm, and sweet fragrance. It is frequently used as a spice and gives food, drinks, and baked items a lovely scent.

C) Cloves: The scent of cloves is powerful, fragrant, and spicy. They are frequently employed in baking and cooking to provide a rich, cozy, and slightly sweet scent to food.

D) Cardamom: This spice has a distinct aroma that is both sweet and somewhat spicy. Cardamom is quite aromatic. With its pleasantly fragrant flavor, it is frequently used in both savory and sweet recipes.

5. Correct option (D)

All of the techniques described - sugar syrups, honey, and refrigeration - may be utilized for fruit preservation. There are several ways to preserve fruits.

Sugar syrups: By submerging fruits in sugar syrups, which helps maintain their flavor, texture, and color, fruits can be preserved. Inhibiting the growth of germs that might contaminate the fruit is how the sugar syrup functions as a preservative. Fruits including cherries, peaches, pears, and pineapples are frequently preserved using this technique.

Honey: Honey may also be used as a natural fruit preserver. By fostering an environment that is unfavorable for microbial development, it aids in preventing spoiling. Honey may be used to preserve fruits by coating them in it or by preparing jams and preserves with it.

Refrigeration: Fruits may be kept in their fresh form for a longer amount of time by using refrigeration. The fruit's shelf life is increased by lowering the temperature since it inhibits microbial development and enzymatic activity. Fruits may be kept fresh and maintained for a number of days or weeks by being stored in the refrigerator, depending on the fruit variety.

6. Correct option (D)

Root: The root of several plants, including carrots, radishes, beets, and potatoes, can be the only edible portion. Because of their nutritional value, these roots are frequently used in many culinary dishes as vegetables.

Stem: Some plants have edible stems that are commonly consumed as food. Examples include asparagus, celery, broccoli, and bamboo shoots. These stems are often cooked or used raw in salads and stir-fries.

Leaves: Many plants contain edible leaves that may be utilized as either vegetables or herbs. The following are some examples: lettuce, spinach, kale, cabbage, basil, and mint. Both raw and cooked meals may be made with these leaves.

Therefore, depending on the particular plant species and the culinary customs of various cultures, the roots, stems, leaves, fruits, and seeds of plants can all be used as food.

7. Correct option (A)

8. Correct option (B)

The staple cuisine of Rajasthan, a desert state, is roti, a sort of flatbread prepared from wheat flour. It is a common dish in Rajasthan and is eaten with a number of other cuisines, including dal, sabzi, and curry.

The flatbread known as chapati is also created from whole wheat flour. Although it is a widespread dish in many regions of India, it is less frequent in Rajasthan than roti.

In Rajasthan, rice is not a common meal. The southern states of India, where the environment is more humid, are where it is more prevalent.

Sambar and idli are South Indian foods. In Rajasthan, they are not frequently consumed.

9. Correct option (D)

Major food crops grown in India include jowar, wheat, and rice.

In India, rice is the most significant crop for food production. Although it is cultivated across the nation, the eastern and southern areas are where it is most significant.

In India, wheat is the second most significant food crop. It is cultivated in the northern and western states since they have more favorable climates for growing wheat.

A kind of millet called jowar is cultivated in the central and western regions. It is crucial in regions with unpredictable rainfall since it is a drought-resistant crop.

Approximately 70% of India's entire food grain production is comprised of these three crops.

10. Correct option (B)

Indian food frequently uses the strong spice asafoetida to add flavour. It belongs to the same family as celery and carrots, the Umbelliferae. It is made from the resinous sap of the roots of Ferula species of plants, mainly Ferula assa-foetida. The plant's thick taproot is used to extract the sap, which is then dried to make resin that is either powdered or utilised in its raw state. Asafoetida is not commonly made from the plant's stem, leaves, or blossoms.

Indian food uses asafoetida, particularly in pickles, curries, and lentil meals. Traditional medicine also uses it to treat colic, indigestion, and gas.

11. Correct option (A & B)

All domestic chickens are members of the genus Gallus, which also contains the red junglefowl. Male chickens are called roosters, while female chickens are called hens. Food, eggs, and feathers may all be obtained from hens and roosters.

Goats are not chickens. They are animals belonging to the Capra genus, which also contains antelopes, sheep, and goats. Normally, goats are not utilized for food, eggs, or feathers.

12. Correct option (D)

We get our minerals from a variety of sources, including plants, animals, and the sea.

Numerous minerals, including calcium, iron, magnesium, potassium, and zinc, are abundant in plants. Leafy green vegetables, legumes, nuts, seeds, and whole grains are a few of the greatest plant-based sources of nutrients.

Animals are a good source of minerals including iron, zinc, selenium, and iodine. Meat, poultry, fish, eggs, and dairy products are some of the richest sources of minerals from animals.

Minerals including iodine, selenium, and magnesium is abundant in seawater. Seafood, seaweed, and salt are a few of the richest sources of minerals found in the ocean.

13. Correct option (A & B)

Both Spinach and coriander plants feature edible leaves that may be used in a number of ways.

Leafy greens like spinach are a good source of vitamins A, C, and K. It is an excellent source of fiber and iron as well. Eaten raw in salads or cooked in a variety of ways, spinach leaves are versatile.

The herb coriander is a common ingredient in international cuisines. Coriander leaves are used to flavor curries, chutneys, and other foods because of their pungent, lemony flavor. Tea and food garnishes may both be made with coriander leaves.

Acacia leaves are rarely used in cooking and are not edible.

14. Correct option (A & B)

Two of the most popular dishes in South India are rice and dal. The primary diet in South India is rice, which is eaten with a variety of foods including dal, sambar, and chutneys. A well-liked food in South India is dal, a lentil soup.

The flatbread known as Phulka is not as popular in South India as rice and dal. Phulka is frequently eaten with curries and other foods and is primarily prepared from wheat flour.

Rice and dal are the two food categories that are most popular in South India. Therefore, D is the correct response.

15. Correct option (D)

16. Correct option (a-Q, b-T, c-R, d-P, e-S)

a) Boiling - Q) Potatoes, Eggs

Food products are cooked by immersion in hot water or another liquid during the boiling process. To get the proper amount of doneness, potatoes and eggs are frequently cooked. As a result, Group A includes potatoes and eggs.

b) Roasting - T) Chicken, fish

Cooking food in an oven or over an open flame while typically employing dry heat is known as roasting. For better flavor and a crispier texture, fish and chicken are frequently roasted. Consequently, Group B includes fish and chicken.

c) Chopping - R) salads

Food products are sliced into smaller, more manageable pieces by the process of "chopping." Salads can contain a wide range of items, but typically, vegetables and other salad elements are prepared by being chopped. As a result, Group B includes salads.

d) Fermentation - P) Bread, Cakes

Sugars are transformed into alcohol or organic acids during the fermentation process by the activity of yeast, bacteria, or other microorganisms. Food products that go through fermentation include bread and cakes, which often use yeast as a leavened. Therefore, Group B includes bread and cakes.

e) Steaming - S) idle

When food is cooked by steam produced by boiling water, it does not come into direct contact with the liquid. Indian traditional food known as idle is prepared by steaming a fermented batter largely comprised of rice and black lentils. Idle is therefore in Group A.

LEVEL-II

1. Correct option (A)

Nutrients, which are helpful components found in food, are crucial for the healthy operation and upkeep of the body. Water, carbs, proteins, lipids, vitamins, and minerals are all considered nutrients. These compounds stimulate development and repair, supply energy, control body functions, and promote general health and wellbeing.

The capacity to perform work is energy. When we eat food, our bodies break down the nutrients into smaller molecules that can be used for energy.

Weight is a unit of measurement for an object's weight. Nutrients don't include weight. It is only a measurement of an object's mass.

Energy-giving nutrients include carbohydrates, which are a type of nutrient. Bread, pasta, grains, fruits, and vegetables, among other foods, contain them.

2. Correct option (B)

The components of food known as nutrients are what our bodies require to remain healthy. They provide us energy, support our physical and mental development, and keep us healthy. Our immune system, which is our body's line of defense against sickness, may be strengthened by certain foods including vitamins and minerals.

A strong immune system is necessary for illness prevention. Our immune system can fend against pathogens and viruses that can make us ill while it is healthy. Vitamin A, C, D, zinc, and selenium are nutrients that support a healthy immune system.

Other nutrients, such as the antioxidants in fruits and vegetables, can aid in lowering the risk of chronic illnesses including diabetes, cancer, and heart disease. As a result, eating a balanced, nutrient-rich diet is crucial for preventing disease and sustaining good health.

3. Correct option (A)

Paddy is the unhusked grain of rice. It is the most common type of rice used in cooking. To cook rice, we need to wash the paddy and then cook it in water. The amount of water we need to use depends on the type of rice we are using. The husk is subsequently removed from these grains during processing, leaving just uncooked rice. When preparing rice, we mix raw rice with water, heat the mixture, and then the rice absorbs the water and cooks.

Wheat, jowar, and rava are not used to cook rice. Wheat is used to make bread, jowar is used to make jowar roti, and rava is used to make rava idle.

4. Correct option (C)

The different items or components used to produce a meal are known as ingredients. They are the precise ingredients or mixtures of ingredients that are mixed together and utilized in a particular recipe or dish in a certain amount or ratio. Among the food elements that affect a dish's flavor, texture, and general make-up are vegetables, fruits, meats, grains, spices, herbs, oils, and other ingredients. They are necessary for cooking and have a significant impact on how good the food turns out.

Items that are not utilized to produce a meal are referred to as non-ingredients. They might be anything from kitchenware and cutlery to cleaning products.

Any object or thing can be referred to by the broad word "items." Items can refer to ingredients, things other than ingredients, or even the meal itself when discussing food.

A prepared meal is specifically referred to as a dish. An ingredient can be used alone or in combination to create a meal.

5. Correct option (A)

The animal's flesh that is consumed as food is known as meat. It contains plenty of iron, protein, and other minerals. Cattle, pigs, lambs, chickens, and fish are just a few of the animals whose meat is available. In many diets, meat is a substantial source of protein, vitamins, minerals, and other nutrients. It is a staple cuisine in many cultures all over the world and is frequently consumed by people as part of their normal diet.

Meat cannot be obtained from plants. Other foods including fruits, vegetables, and grains can be found in them as well.

6. Correct option (D)

Plants are the source of fruits and vegetables. Vegetables include a variety of edible plant components such leaves, stems, roots, and bulbs, whereas fruits are the mature ovaries of blooming plants and carry seeds. They are abundant sources of fiber, vitamins, minerals, and other healthy substances. Humans frequently consume fruits and vegetables because of their nutritional worth and health advantages. They are also an essential component of a balanced diet.

Options A) Oil and Ghee, and B) Milk and Pulses, and C) Ghee and Milk are incorrect because they include animal-derived products (oil and ghee from animals, milk from animals, and pulses being a plant-based source of protein).

7. Correct option (A)

Both the meat and the ghee come from animals. Ghee, a sort of clarified butter that is commonly used in cooking, is produced from butter that has been boiled and the milk particles taken out. Meat is any animal's edible flesh, including that of fish, fowl, pigs, and cattle. These foods, which come from animal origins, are used as food in many cultures and provide nutrients including protein, fats, vitamins, and minerals.

Option B) Fruits and Vegetables and Option C) Dry fruits and Ghee are incorrect because fruits, vegetables, and dry fruits are obtained from plants, not animals.

8. Correct option (D)

Cereals are the grains of certain grasses that are used for food. They are a good source of carbohydrates, fiber, and vitamins. Some common cereals include rice, wheat, maize, millets, barley, and oats.

The two grains that are most extensively grown worldwide are rice and wheat. For many people all across the world, they are a staple meal. Another important cereal crop is maize, which is used to produce a range of goods including cornmeal, corn flour and corn oil. Small, hard-grained grains known as millets are cultivated in warm areas. They are frequently used to produce porridge and bread, and they are a rich source of protein and fiber. Beer, bread, and breakfast cereals all use the cereal barley. Making muesli and muesli using oats, a grain rich in protein and fiber, is quite common.

9. Correct option (A & B)

Both coconut and mustard seeds may be used to make oil.

A) Mustard, Coconut - Mustard oil, which is extensively used in cooking and other culinary purposes, is typically extracted from mustard seeds. Coconut oil, which is frequently used in baking, cooking, and cosmetics, is extracted from coconut seeds.

B) Coconuts, sesame - As previously indicated, coconuts are utilized to obtain coconut oil. Sesame oil, sometimes referred to as gingelly oil or sesame oil, is extracted from sesame seeds and used in cooking, salad dressing, and flavoring.
C) Guava, Sunflower, on the other hand, is inaccurate because guavas are fruits and aren't normally utilized to produce oil. However, sunflower oil is frequently extracted from sunflower seeds.

10. Correct option (D)

Spices are indeed used for various purposes, including flavoring, coloring, and preservation.
a) Flavoring: Spices are widely used to enhance the taste, aroma, and overall flavor profile of dishes. They add depth, complexity, and distinct flavors to culinary preparations.
b) Coloring: Some spices, such as turmeric, paprika, saffron, and annatto, are used to provide natural coloring to food. These spices can impart vibrant hues and enhance the visual appeal of dishes.
c) Preservation: Certain spices possess antimicrobial properties that can help inhibit the growth of harmful bacteria or microorganisms. As a result, they are used to some extent in food preservation to extend shelf life or prevent spoilage.

11. Correct option (A)

The seeds of the Cuminum cyminum plant are dried and used to make the spice cumin. It is a Mediterranean and Middle Eastern native and a member of the parsley family. Indian, Middle Eastern, and Mexican cuisine frequently employ cumin because of its powerful, earthy flavor. Additionally, it's a component of certain stews and sausages from Europe.
Spices are not sugarcane or jaggery. A plant called sugarcane is used to produce sugar. Sugar that is manufactured from sugarcane is called jaggery. In contrast to spices, they are not used to flavor food.

12. Correct option (C)

The leaves of the tea plant, Camellia sinensis, are used to make tea. To make several kinds of tea, including black tea, green tea, oolong tea, and white tea, the young leaves and leaf buds of the tea plant are plucked and processed. To create their distinctive flavors, the leaves are dried and put through a variety of oxidation and fermentation procedures.
On the other hand, coffee is made from coffee beans, which are the Coffea plant's seeds. The fruit, which is frequently referred to as coffee cherries, contains the coffee beans. To bring out the distinctive flavors and fragrances of coffee, coffee cherries are selected, and then the seeds are removed and roasted.
Therefore, the correct answer is C) Leaves and Seeds - Tea is obtained from the leaves of the tea plant, and coffee is prepared from coffee seeds (beans).

13. Correct option (B)

Plants prepare their own food through the process of photosynthesis. Photosynthesis is a biochemical process that occurs in the chloroplasts of plant cells, primarily in the leaves. It involves the conversion of light energy from the sun into chemical energy in the form of glucose (sugar). During photosynthesis, plants use carbon dioxide from the atmosphere, water from the roots, and sunlight to produce glucose and oxygen.
When an organism breathes, oxygen is used to break down glucose and release energy. All live cells go through this procedure.
The process through which plants lose water via their leaves is called transpiration. This procedure circulates water and nutrients throughout the plant while also aiding in cooling it.
The process through which blood is circulated around the body is called circulation. This procedure is necessary for delivering nutrients and oxygen to the cells as well as for eliminating waste.

14. Correct option (C)

Regarding food chains and ecosystems, plants are viewed as producers. Through the process of photosynthesis, as was previously established, they have the capacity to produce their own food. Producers are living things that can transform light energy into chemical energy by using sunlight and inorganic elements (such carbon dioxide and water) to create organic molecules, mainly glucose. Other creatures in the food chain can obtain energy from these chemical molecules.

Consumers are living things that can't cook for themselves. To obtain the necessary energy, they must consume other living things. Herbivores, carnivores, and omnivores are all types of consumers. Creatures that decompose dead creatures are called decomposers. In order for producers to utilize the nutrients from the deceased organisms, they release them back into the environment.

Therefore, plants are only producers.

15. Correct option (A)

Plants store food primarily in the form of starch. Starch is a complex carbohydrate made up of glucose units that are linked together. It serves as the main energy storage molecule in plants. During photosynthesis, excess glucose produced is converted into starch and stored in various plant parts, such as roots, tubers, seeds, and fruits. Starch is insoluble in water, which makes it an efficient and compact storage form for plants to store energy for future use.

Although plants do contain proteins, they are not the principal food storage type in plants, hence Option B) is wrong.

Option C) is false because, although being a mineral necessary for plant growth and development, calcium is not the major way that plants store food.

16. Correct option (D)

Carrots are a type of root vegetable. Plants that produce root vegetables store nourishment in their roots. The carrot root is a taproot, which means it descends directly into the ground. The carrot plant's primary storage organ is the taproot.

The root system of the carrot plant stores nutrients and energy, enabling it to develop and survive under unfavourable conditions.

Because of its sweet and crisp flavour and nutritional content, the carrot plant's root is th e component that is frequently eaten as a vegetable. It holds vitamins, proteins, and carbs.

Carrot leaves have a green, fluffy appearance. The carrot root's crown is where they are connected. The leaves assist the carrot plant's photosynthetic process and food production.

17. Correct option (A)

Apples are a type of plant's fruit. The mature ovary of the apple tree's blossom is what we know as an apple. It grows from the fertilized blossom and contains apple tree seeds. Apples come in a variety of colors, shapes, and flavors. The apple fruit is a good source of vitamins and minerals. It is also a good source of fiber. Apples are a popular fruit that can be eaten fresh, cooked, or juiced.

Option B) Leaf is wrong since it is not the portion of the plant that produces the apple fruit; rather, it is a distinct plant component engaged in photosynthesis and gas exchange.

Option C) Because apples are officially categorized as fruits and not leaves, Options A and B are equally erroneous.

18. Correct option (A)

The edible portion of the black gramme plant is the seed, which is a sort of pulse or legume. Small, oval-shaped legumes known as black gramme seeds, urad dal, or black lentils are frequently used in a variety of dishes around the world. Black gramme seeds are an

important component of vegetarian and vegan diets because they are high in protein, fiber, and other minerals.

Black gram is not a fruit, hence Option B) Fruit is wrong.

Both Options C) Leaf and D) Root are inaccurate since they do not depict the section of the black gram plant that is edible.

19. Correct option (A)

Milk is primarily obtained from cows and is commonly consumed as a beverage. Cows are widely domesticated for milk production, and their milk is a rich source of essential nutrients such as protein, calcium, vitamins, and minerals. Milk is consumed in its natural form or used as an ingredient in various dairy products such as cheese, yogurt, butter, and cream. While there are other animals that produce milk, such as goats, sheep, and buffaloes, the most commonly consumed and widely available milk is derived from cows.

Option A) Cat and Option C) Cat are inaccurate because, although cats may produce milk to feed their kittens, humans do not normally ingest it on a daily basis.

20. Correct option (B)

Rearing and management of fish on a large scale is known as pisciculture. Pisciculture is the practice of breeding, rearing, and cultivating fish in ponds, tanks, or other controlled aquatic environments. It involves activities such as stocking fish, providing suitable conditions for their growth and reproduction, managing water quality, feeding, and harvesting. Pisciculture is an important practice for commercial fish production, aquaculture, and meeting the demand for fish as a food source.

The other options are incorrect. Poultry is the raising of domesticated birds for food. Domestication is the process of taming and breeding animals for human use. Agriculture is the practice of cultivating plants and livestock.

21. Correct option (D)

Poultry is the general term for domesticated birds such hens, geese, quails, and ducks. In poultry farming, the birds are raised and managed for a variety of uses, including the production of eggs and meat (poultry meat). The agriculture sector's key sector is poultry farming, which is crucial in supplying the world's need for chicken products. Poultry is a major source of protein for humans. It is also a good source of other nutrients, such as vitamins, minerals, and iron.

The other options are incorrect. Pisciculture is the practice of rearing and managing fish on a large scale. Agriculture is the practice of cultivating plants and livestock. Dairy Farm is a place where dairy cows are raised for milk.

22. Correct option (C)

A cow is an example of an herbivorous animal. Animals known as herbivores are those that eat predominantly plant-based diets including grass, leaves, and other types of flora. Cows' specialized digestive systems, which include many chambers in the stomach, enable them to effectively digest and extract nutrients from plant matter. They have developed to rely on an herbivorous diet and can feed on grass and other plant materials with ease.

Carnivores are animals that eat meat. They have sharp teeth and claws that help them to catch and kill their prey. Examples of carnivores include lions, tigers, and wolves.

Omnivores are animals that eat both plants and meat. They have a variety of teeth and digestive systems that allow them to eat both types of food. Examples of omnivores include humans, bears, and pigs.

23. Correct option (B)

Evaporation is the method used to produce salt from seawater. Various dissolved salts may be found in sea water, with sodium chloride (NaCl) being the most prevalent. Sea water is gathered and allowed to evaporate to produce salt, either naturally through solar

evaporation or artificially. The leftover salt crystals are gathered when the water evaporates, concentrating the salt content. Salt is a valuable commodity that has been used by humans for centuries. It is used in cooking, preserving food, and making soap and other products.
Option A) pure water is wrong since it is uncommon to extract salt from sources of pure water. Compared to sea water, fresh water sources have lower salt content.
Option C) Polluted water is inaccurate as well since it is not a reliable or appropriate supply of salt.

24. Correct option (B)

Omnivores are organisms that have the ability to consume and derive nutrients from both plant-based and animal-based sources
Human and Cow - Humans are omnivorous, eating both plant and animal foods in their diets, but cows are herbivorous, eating mostly plant-based meals. This choice is thus unsuitable.
The Human and the Crow – Both people and crows are omnivorous. While crows are opportunistic feeders who eat a range of items, including fruits, seeds, insects, and small animals, human diets are broad and contain both plant and animal foods. So, this choice is the right one.

25. Correct option (A)

Animals that feed on the flesh of dead animals are called scavengers. Scavengers play an important ecological role by consuming and disposing of decaying organic matter. They help to clean up the environment by recycling nutrients from dead animals back into the ecosystem. Examples of scavengers include vultures, hyenas, crows, and certain species of beetles.
Option B) Parasites is incorrect because parasites are organisms that live in or on another organism, called the host, and obtain nutrients from the host's body. Parasites may feed on the host's tissues or body fluids, but they are not specifically associated with feeding on the flesh of dead animals.
Therefore, the correct answer is C) Only A - The animals that feed on the flesh of dead animals are called scavengers.

26. Correct option (B)

An insect parasite is a mosquito. In particular, female mosquitoes are hematophagous, meaning they consume other creatures' blood, including human blood. They penetrate the skin with their unique mouthparts to draw blood, which provides nutrients for the development of the egg. Female mosquitoes need a blood meal in order to lay eggs; male mosquitoes, on the other hand, mostly consume nectar and plant liquids.
Option A) Scavengers is inaccurate since mosquitoes do not feed on decaying organic debris or dead animals, whereas scavengers do.
Option C) Decomposers is inaccurate since decomposers are creatures that help in the decomposition process by breaking down dead organic matter into simpler forms. Mosquitoes don't contribute much to decomposition.
Because they feed on the blood of other creatures, mosquitoes are regarded as parasites, hence the correct response is B).

27. Correct option (A)

Endoparasites are living things that reside inside another creature. They can lead to a number of disorders and are frequently tiny. Endoparasites are parasites that reside within the body of their host. They obtain their nutrition and other resources by feeding on the tissues, fluids, or blood of the host. Tapeworms, roundworms, and parasites that cause malaria are a few examples of endoparasites.

Ectoparasites, in contrast, exist outside of another organism's body. They may also spread illness and are frequently bigger than endoparasites. Ectoparasites include things like fleas, ticks, and lice.

Organisms that decompose dead organic substances are called decomposers. They return nutrients to the environment, where other creatures can utilize them. Among the decomposers are fungus and bacteria.

Organisms that can produce their own nourishment are known as producers. They produce molecules that are high in energy, like glucose, using sunlight, water, and carbon dioxide. Examples of producers include some bacteria, algae, and plants.

28. Correct option (A)

Peacocks are indeed omnivorous birds, meaning they consume both plants and animals. While their diet consists mainly of plant matter such as seeds, fruits, and grains, they also opportunistically feed on insects, small reptiles, and other small animals. Peacocks are known to display a varied diet, making them omnivores.

The first option, "herbivorous," is untrue because peacocks eat a variety of foods. They are omnivores because they eat both insects and small animals in addition to plant-based meals like seeds and fruits, making them omnivores.

Option B) Carnivorous: This choice is untrue since peacocks do not eat meat or flesh as their main source of food. Even though they occasionally eat small animals, they do not eat primarily carnivorous foods.

Producer (Option D): This choice is untrue since producers are living things with the capacity to grow their own food through processes like photosynthesis. Being animals, peacocks must rely on outside food sources because they are unable to manufacture their own food.

29. Correct option (B)

Through mechanisms like photosynthesis, animals are unable to create their own nourishment. They rely on eating other living things or organic material to get the food and energy they require to survive. As a result, in terms of ecology, animals are considered consumers. species that consume other species or their products are known as consumers. They get their energy from the food they eat, which they also digest.

Producing your own food through processes like photosynthesis or chemosynthesis requires creatures like plants, algae, or certain bacteria, which are not producers. Therefore, Option A) Producer is false.

Option C) Decomposer is inaccurate since decomposers are creatures that help in the decomposition process by breaking down dead organic matter into simpler components.

Option D) None is inaccurate because it is true that animals are considered to be consumers.

30. Correct option (B)

From poultry, which refers to domesticated birds like chickens, we obtain eggs and meat.

Eggs are a typical and popular dietary item made from fowl. Eggs are harvested from chickens that have laid eggs for human use. Eggs are adaptable and utilized in a variety of culinary dishes.

Another thing you may get from fowl is meat. The flesh of chickens and other poultry animals is grown for human consumption. In many cuisines, poultry meat—such as chicken meat—is a substantial source of protein.

Option A is untrue since milk is not produced by using chickens. Most animals, including cows, goats, and sheep, provide milk.

Option C) is untrue since honey is not produced by poultry, which makes egg and honey inaccurate. Honey is created by honey bees from floral nectar.

Option D) Egg and fish is incorrect because fish is not obtained from poultry. Fish is a separate category of aquatic animals, not related to poultry.

2. Components of Food
Solutions

LEVEL-I

1. Correct option (C)

The numerous nutrients included in food ingredients are substances that our bodies require for normal operation and good health. Carbohydrates, proteins, lipids, vitamins, minerals, and water are all examples of nutrients. Each nutrient performs a specialized function in the body, such as supplying energy, boosting the immune system, repairing and developing tissues, and preserving numerous physiological functions.

Carbohydrates, proteins, and fats are all nutrients. However, they are not the only nutrients that the body needs. Other important nutrients include vitamins, minerals, and fiber.

2. Correct option (D)

The common forms of carbohydrates found in food are starch and sugars.

A) Starch: Starch is a type of complex carbohydrate that is present in foods such as grains (such as rice and wheat), tubers (such as potatoes), and legumes (such as beans and lentils). It is a kind of energy storage in plants made up of extended chains of glucose molecules joined together.

B) Sugars: Sugars are simple carbohydrates that come in a variety of forms, including fructose from fruits, sucrose from table sugar, lactose from milk, and honey. They are made up of one or two sugar molecules and offer the body a rapid source of energy.

C) Glucose: A kind of sugar, glucose is not the most prevalent form of carbohydrates present in meals, despite being the body's main source of energy.

3. Correct option (B)

Proteins are regarded as nutrients that help build muscle since they are essential for the development, maintenance, and repair of bodily tissues. Amino acids, the body's building components, make up what they are. The growth and repair of muscles, organs, skin, hair, nails, and other bodily tissues depend on proteins.

Although they may create energy, proteins do not have this as their main purpose. The primary elements that provide the body energy are carbs and fats. Proteins are crucial for general growth and health because they play a major role in structural and functional activities. So, "Body-building foods," option B, is the appropriate response.

4. Correct option (A)

The Biuret test, which is used to check for the presence of proteins, is the procedure you described. A food item that includes proteins becomes violet when copper sulphate and sodium hydroxide are added to it. This is as a result of the proteins' reaction with copper sulphate to create a complex that has a violet color.

When added to the test solution, fats and starches will not cause the test solution to become violet because they do not react with copper sulphate and sodium hydroxide.

5. Correct option (C)

A balanced diet is one that has all the necessary nutrients in the appropriate amounts. This implies that it has the proper amounts of carbs, lipids, proteins, vitamins, minerals, and roughage.

The body needs all of the key nutrients—carbohydrates, lipids, proteins, vitamins, and minerals—for effective operation and general health.

Proteins, lipids, and carbohydrates are macronutrients that support diverse physiological functions and provide us energy.

Micronutrients like vitamins and minerals are necessary in lesser amounts but are just as crucial for many body processes. These processes include supporting metabolism, preserving a robust immune system, and ensuring that enzymes and hormones work as intended.

The two categories of micronutrients that are listed in Option C, "Vitamins & Minerals," are accurately identified.

6. Correct option (C)

A balanced diet is one that has all the necessary nutrients in the appropriate amounts. This implies that it has the proper amounts of carbs, lipids, proteins, vitamins, minerals, and roughage.

For the purpose of sustaining excellent health, a balanced diet is crucial. Numerous health issues, including as obesity, heart disease, stroke, and cancer, can be avoided with its aid. Energy levels, happiness, and general wellbeing can all be helped by it.

When the body has too much fat, the condition is called obesity. It results from consuming too many calories and not working out enough.

A diet that lacks all the necessary nutrients in the proper amounts is considered deficient. This may result in a number of health issues, such as weariness, weakness,

7. Correct option (A)

A form of fiber known as roughage is not digested by the body. It can help lower the risk of constipation and maintain the health of the digestive system. By giving the stool more weight, roughage aids in the removal of undigested food from the body by causing the intestinal walls to contract. Food is pushed through the digestive system by this motion.

The body need vitamins, minerals, and proteins among other vital components to function effectively. They are not, however, broken down by the body the same way as roughage is. The small intestine is where vitamins, minerals, and proteins are absorbed into the circulation. Roughage does not enter the circulation; rather, it travels through the digestive system and is expelled from the body.

8. Correct option (B)

Thiamine (vitamin B1) insufficiency leads to the illness beriberi. This is because thiamine is essential for the proper function of the nervous system and muscles. When the body does not get enough thiamine, the muscles can become weak and fatigued. It mostly affects the neural system and can cause a variety of symptoms, such as muscular weakness and atrophy, trouble walking, and problems with the nerves. The word "beriberi" is derived from the Sinhalese word for "weakness."

9. Correct option (C)

Although minerals are necessary nutrients for the body, they are only required in modest amounts when compared to macronutrients like carbs, proteins, and fats. Minerals are essential for a number of physiological activities, including bone health, neuron function, enzyme activity, and maintaining a healthy fluid balance, despite the fact that their requirements are quite low.

The number of minerals required for the body's optimal operation is best summed up by Option C, "Small amounts." Like macronutrients, minerals are not required in huge quantities, but their presence is essential to maintain general health and wellbeing.

10. Correct option (A)

Carbohydrates and fats are both macronutrients that serve as a significant source of energy for the body. They are crucial for preserving a healthy weight since they are the body's primary source of energy. Glucose, the body's primary energy source, is created when carbohydrates are broken down. Fatty acids are created when fats are broken down, and they may either be utilized right away for energy or stored for future use.

Carbohydrates are more easily and quickly converted into energy, while fats provide a more sustained and longer-lasting source of energy.
Protein-rich diets are foods that are good for growing muscle. Building and repairing bodily tissues depend on protein. Foods that demand energy for digestion are called energy consumers. High-protein foods and foods high in fiber are among these. Foods that are protective contain vitamins, minerals, and other elements that guard the body from illness. These include whole grains, fruits, and vegetables.

11. Correct option (D)

Diseases associated with vitamin deficiencies are brought on by a sustained shortage of certain nutrients in the diet. Numerous deficiency disorders can develop when the body doesn't get enough important nutrients for an extended length of time, such as vitamins, minerals, proteins, or lipids. Each vitamin is essential to the body's operation, and shortages or particular health problems can occur from its absence or insufficient ingestion. For instance, a shortage of vitamin C can result in scurvy, a lack of iron can result in anemia, and a lack of calcium can cause brittle bones and osteoporosis. To avoid deficiency disorders, it is crucial to maintain a balanced diet with a range of foods.

12. Correct option (D)

A food item that includes starch turns blue-black when iodine solution is added. This is due to the blue-black complex that is created when starch and iodine react.
Protein and glucose do not react with iodine solution, therefore adding them does not cause them to become blue-black. So, you may add a few drops of iodine solution to a food item to check for the presence of starch. The food item includes starch if the color turns to blue-black.
Glucose and protein do not react with iodine solution, so they will not turn blue-black when added to iodine solution.

13. Correct option (C)

Minerals include calcium, phosphorus, and iron. The body requires trace amounts of minerals, which are inorganic elements, to carry out a variety of physiological processes. They are crucial for sustaining general health since they are involved in various activities like iron transfer, enzyme activity, bone health (calcium and phosphorus), and oxygen transport. Although the body also needs vitamins, carbs, and roughage (fiber), the ideal response in this situation is C) Minerals.

14. Correct option (A)

Numerous nutrients, such as iron, zinc, magnesium, and selenium, are abundant in meat. These minerals are necessary for a number of body processes, including the creation of blood cells, energy synthesis, and immunological response.
Although they may not have as much diversity of minerals as meat, rice, wheat, and bananas are nonetheless excellent providers of several minerals.

15. Correct option (B)

The body requires a lot of nutrients called macronutrients. They consist of proteins, lipids, and carbs. Proteins and fats are necessary for the body's normal operation. They give out energy, support tissue growth and repair, and safeguard organs. Carbohydrates, proteins, fats and water are macronutrients. Vitamins, minerals, roughage are micronutrients.
Here are some of the functions of fats and proteins:
Fats:

- Provide energy
- Help to build and repair tissues
- Protect organs
- Keep the body warm

- Help to absorb vitamins

Proteins:

- Provide energy
- Help to build and repair tissues
- Make enzymes, hormones, and antibodies
- Transport nutrients throughout the body
- Help to regulate blood sugar levels

So, in the given options B) Fats and proteins are the right option in which both are macronutrients.

16. Correct option (C)

Iodine deficiency leads to goiter. The thyroid gland, a butterfly-shaped gland in the neck, enlarges in a condition called goiter. Iodine is a crucial element required for the synthesis of thyroid hormones. The thyroid gland enlarges in an effort to generate more hormones when the body is deficient in iodine, which results in the development of goiter. Iodine supplementation and iodized salt programmers can avoid this condition, which previously was common in areas with low iodine levels in the soil and water.

A is incorrect because iron deficiency does not cause goiter. B is incorrect because silver deficiency does not cause goiter. D is incorrect because salt does not contain iodine.

17. Correct option (B & C)

Due to its crucial function in preserving the health and integrity of these tissues, calcium shortage can lead to weak bones and tooth decay.

i) Tooth decay

Calcium is a necessary mineral that is required for the upkeep of sturdy and healthy teeth. It contributes to the development of the tooth's hard structure, or enamel, which guards the inner layers from acids and bacteria that cause decay. When there is a calcium deficit, the enamel weakens and is more prone to erosion, which causes tooth decay.

ii) Weak bones:

Another important element of bone formation is calcium. 99% of the calcium in the body is kept in the bones and teeth. The body may use calcium from the bones to maintain normal blood calcium levels when there is insufficient calcium intake. This can weaken the bones over time and result in diseases like osteoporosis, increasing the risk of fractures and other accidents.

18. Correct option (D)

Vitamins are referred to as "protective foods" since they are crucial for sustaining general health and shielding the body from illnesses and deficits. They are organic substances that the body needs in little amounts for a variety of physiological processes. Vitamins function as coenzymes, helping enzymes complete crucial chemical processes.

Option A, "Foods that produce energy," and Option C, "Foods that consume energy," are inaccurate definitions of vitamins. Foods that provide energy often relate to macronutrients (carbohydrates, proteins, and fats) that supply calories for energy generation.

Option B (Body-building foods) might refer to proteins, which are macronutrients responsible for building and repairing tissues in the body. While vitamins are essential for various bodily functions, they are not primarily known as body-building foods.

19. Correct option (A)

Vitamins and minerals known as micronutrients are critical for human health. Even though they are just somewhat necessary, they are essential for numerous body processes.

Organic substances known as vitamins are crucial for development, growth, and overall health. Vitamin C and B are examples of water-soluble vitamins, and the other category includes fat-soluble vitamins. vitamin A, D, E, and K

Minerals are inorganic substances that are vital to maintaining good health. They are essential for several biological processes, such as neuron, bone, and blood clotting health. Minerals including calcium, iron, magnesium, phosphorus, potassium, and zinc are among the most crucial ones.

20. Correct option (B)

Scurvy is a condition brought on by a deficiency of vitamin C. Collagen, a protein that provides skin, bones, and blood vessels their strength, is formed only when vitamin C is present. Additionally, it aids in immune system stimulation and infection prevention.

The gums swell and bleed readily when the body is deficient in vitamin C. Additionally, wound healing takes longer. Scurvy also causes joint discomfort, muscular weakness, and weariness.

The illness rickets is brought on by a deficiency of vitamin D. When the body doesn't produce enough healthy red blood cells, it has anemia. Any of these disorders do not manifest as visual loss.

21. Correct option (D)

Our bodies require a balanced diet in order to be strong and healthy. Consuming a range of meals that contain all the necessary nutrients, such as carbs, proteins, fats, vitamins, minerals, and water, is the definition of a balanced diet. Each vitamin supports body processes, cellular development, and general health in a different way.

While certain items like rice (option A) and meat (option B) can be included in a balanced diet, relying only on one or a few particular meals could not provide the body with all the essential elements. Our bodies get the nutrients they need for good health when we eat a variety of meals from various dietary categories. Additionally, it's crucial for a variety of body processes and general health to be appropriately hydrated with water (option C).

22. Correct option (C)

Nutrients are the components that make up various food products. The body need nutrients in order to function effectively. They are broken down into six primary categories: water, carbs, proteins, fats, vitamins, and minerals.

The body's primary fuel source is carbohydrate. They may be found in meals including fruits, vegetables, bread, pasta, rice, and pasta.

The development, maintenance, and repair of the body's tissues depend on proteins. Foods including meat, chicken, fish, eggs, dairy products, and legumes contain them.

For the body to store energy, absorb nutrients, and safeguard organs, fats are necessary. They can be found in foods including nuts, seeds, dairy, meat, chicken, and fish.

Numerous body processes, including eyesight, development, and reproduction, depend on vitamins. Fruits, vegetables, and some animal products all contain them.

Numerous body processes, such as bone health, blood coagulation, and neuron function, depend on minerals. Fruits, vegetables, and some animal products all contain them.

Life is dependent on water. It facilitates the movement of nutrients throughout the body, the elimination of waste, and the control of body temperature.

23. Correct option (C)

When the body doesn't produce enough healthy red blood cells, it has anemia. Because red blood cells are responsible for delivering oxygen to the body's tissues, when there are not enough of them, the tissues do not receive adequate oxygen. Numerous symptoms, such as exhaustion, shortness of breath, pallor, and a racing heart, may result from this.

Anemia commonly manifests as fatigue and a lack of appetite. Pale skin is another typical symptom, but it can also be brought on by other illnesses.

Vitamin D deficiency is the root cause of rickets. Scurvy is a disease brought on by a deficiency of vitamin C. Vision loss is not a sign of anemia.

24. Correct option (D)

Vitamins that dissolve in fat are known as fat-soluble vitamins. They don't need to be refilled as frequently as water-soluble vitamins since they are kept in the body's fatty tissues. The fat-soluble vitamins include A, D, E, and K.

- Vitamin A is essential for vision, growth, and reproduction.
- Vitamin D is essential for bone health and immune function.
- Vitamin E is a powerful antioxidant that protects cells from damage.
- Vitamin K is essential for blood clotting.

25. Correct option (C)

Vitamins that dissolve in water are known as water-soluble vitamins. They must be replaced often via diet as the body cannot store them.

Vitamin C and the B vitamins are examples of water-soluble vitamins.

Collagen, a protein that provides skin, bones, and blood vessels their strength, is formed only when vitamin C is present.

The creation of energy, metabolism, and nerve function are just a few of the crucial body processes that the B vitamins play a role in.

On the other hand, vitamins that dissolve in fat are known as fat-soluble vitamins. The fat-soluble vitamins include A, D, E, and K.

Given that both vitamins B and C are water-soluble, the correct response is C.

26. Correct option (B & C)

Dietary fibers and water are both non-nutrients. They don't provide the body any calories or necessary nutrients. They can, nonetheless, nevertheless be healthy in other ways.

Water helps the body stay hydrated and is necessary for survival. Additionally, it facilitates the movement of nutrients and waste materials throughout the body. Dietary fibers support a healthy digestive tract and may lower the chance of developing certain chronic illnesses.

While lemons are a wonderful source of potassium, vitamin C, and other minerals. They also have dietary fibers in them. Lemons thus do contain nutrients.

27. Correct option (A)

A fat-soluble vitamin called vitamin D is necessary for a variety of body processes, including cell development, immune system health, and bone health. In the presence of sunshine, the body may produce vitamin D. The chemical transformation of a precursor molecule into vitamin D3 takes place when sunlight strikes the skin. Following absorption into the circulation and passage to the liver, vitamin D3 is transformed into calcidiol. The subsequent transformation of calcidiol into calcitriol, the active form of vitamin D, occurs.

28. Correct option (B)

Water aids in the digestion of meals, moves nutrients throughout the body, and maintains the digestive system's health, all of which aid in the body's ability to absorb nutrients.

Our body's ability to absorb nutrients is greatly aided by water. Many nutrients dissolve in water and are known as water-soluble nutrients. Water aids in the digestion of these nutrients and makes it easier for the small intestine to absorb them. Water also aids in the bloodstream's role in the distribution of nutrients throughout the body. Water is therefore necessary for healthy nutrition absorption and for maintaining general body processes.

Without adequate water, our bodies are less able to absorb nutrients, which can result in vitamin shortages.

29. Correct option (B)

a. Night blindness; iii. Poor vision in dim light.

Vitamin A is necessary for eyesight and its lack results in night blindness. Poor vision in low light is one of the signs of night blindness.

b. Beriberi; iv. Poor vision in dim light Patients.
Extreme weakness and a sour tongue are symptoms of beriberi.
Thiamine, a nutrient required for the creation of energy, is deficient in beriberi. Extreme weakness is one of the beriberi symptoms. Another sign of beriberi is a sour tongue, albeit it does not occur as frequently as acute weakness.
c. Goiter; I. Enlarged thyroid gland.
Iodine is required for the synthesis of thyroid hormones and its absence leads to goiter. An enlarged thyroid gland is one sign of goiter.
d. Anemia; ii. Deficiency of hemoglobin
Hemoglobin deficiency is the third kind of anemia. A lack of hemoglobin, a protein that transports oxygen in the blood, results in anemia. A lack of hemoglobin is one of the signs of anemia.
e. rickets; v. Brittle bones in children
Children with rickets have brittle bones. A lack of vitamin D, which is necessary for calcium absorption, results in rickets. Having brittle bones in youngsters is one of the signs of rickets.

LEVEL-II

1. Correct option (A)
A renowned research centre in India that specializes in the study of plant fossils and related areas is the Birbal Sahni Institute of Palaeobotany (BSIP). In the scientific field of paleobotany, plant fossils discovered in rocks, sediments, and other geological formations are identified, examined, and interpreted.
The institution was founded in 1946 and is named for Professor Birbal Sahni, a well-known Indian paleobotanist who made substantial contributions to the discipline. It is located in Lucknow, Uttar Pradesh, India.

2. Correct option (B)
Nutrients are a general term for the chemical elements found in food that give energy and are necessary for bodily growth and repair. The maintenance of several physiological systems and the promotion of general health depend on nutrients. They consist of water, vitamins, minerals, proteins, fats, carbs, and proteins, all of which have different functions within the body. These nutrients, which are required for optimal functioning and wellbeing, can only be acquired from the meals we consume.
Nutrients include things like proteins, carbs, and vitamins. They are not the only kind of nutrient, though.

3. Correct option (D)
Organic substances known as carbohydrates are made up of carbon, hydrogen, and oxygen atoms. Carbohydrates have the generic formula $(CH_2O)n$, where "n" denotes the molecule's number of carbon atoms. The ratio of carbon, hydrogen, and oxygen in carbohydrates is represented by this formula, which has the same number of hydrogen atoms as carbon atoms and half the amount of oxygen atoms. An essential part of carbohydrates, oxygen is necessary to both their structure and function.
Option (A), helium, is not an element found in carbohydrates. Helium is a noble gas that is not chemically reactive.
Option (B), lithium, is not an element found in carbohydrates. Lithium is a metal that is not chemically reactive.
Option (C), nitrogen, is found in some carbohydrates, but it is not the only element found in carbohydrates

4. Correct option (A)

Carbohydrates that you consume in excess of what your body requires will be stored as fat. Obesity and weight gain may result from this. A balanced diet that contains a range of foods, such as carbs, proteins, and fats, is crucial. You may do this to maintain a healthy weight and reduce your chance of developing obesity-related diseases.

Fats cannot be formed from proteins. The building components of biological tissues are called proteins. Using glucose as a fuel source, carbohydrates are converted to energy. Vitamins do not become fat-soluble. The body need vitamins as necessary elements for healthy health.

5. Correct option (A & B)

Both mangoes and rice are abundant in carbs. The body's primary fuel source is carbohydrate. They may be found in a wide variety of foods, such as cereals, fruits, vegetables, dairy products, and other foods.

Complex carbs may be found in rice, a grain. The body breaks down complex carbs gradually, releasing energy gradually. Fruits like the mango are a good source of simple carbs. Simple carbs provide the body a fast energy boost since they are easily broken down by the body.

Both rice and mango are nutritious foods that may be included in a diet that is balanced.

6. Correct option (C)

Amino acids, which are proteins' building components, make up proteins. Proteins are created by combining the 20 distinct amino acids. An amino group (-NH2) and a carboxyl group (-COOH) are chemical molecules that are linked to a central carbon atom to form amino acids. The structure and purpose of a protein are determined by the amino acid sequence.

Carbohydrates, or option (A), are not the constituent parts of proteins. Carbon, hydrogen, and oxygen make up the organic substances known as carbohydrates.

Fats in option (B) are not the fundamental components of proteins either. They are necessary for the growth and development of the body. Vitamins, option (D), are not the components that make up proteins.

7. Correct option (D)

Protein-rich diets are sometimes referred to as "bodybuilding" foods since they are essential for the body's repair and tissue growth processes. Our bodies convert the proteins in diets high in protein into amino acids, which are the building blocks needed for tissue development and repair. The phrase "bodybuilding" meals refers to how crucial this is for maintaining and growing muscles. In addition to producing enzymes and supporting the immune system and hormone synthesis, proteins also perform other vital bodily tasks.

Although option (A), "tissue-building," is a reasonable way to describe how proteins work, it is not the phrase used to specifically characterize foods that are high in protein.

Protein activity cannot be adequately described as energy-giving in Option (B). The body does not primarily obtain its energy from proteins.

The chemical constituents of proteins are known as amino acids, option (C). But amino acids aren't edible.

8. Correct option (A)

Protein is found in meat, milk, eggs, and pulses. For the development and repair of tissues, protein is a crucial nutrient. The synthesis of enzymes, hormones, and antibodies depends on it as well.

A good source of plant-based protein is pulses. They are an excellent source of iron, fiber, and folate as well. An excellent source of animal-based protein is milk. Additionally, it is a good source of potassium, calcium, and vitamin D. Eggs are a rich source of fat and protein

from animal sources. They are also an excellent source of selenium, choline, and vitamin B12. An excellent source of animal-based protein is meat. Additionally, it is an excellent source of zinc, vitamin B12, and iron.

9. Correct option (D)

Rich sources of fat include ghee, butter, and oil. Fatty acids, which are chains of carbon atoms connected by bonds, are the building blocks of all of them. When the body breaks down the fat, a lot of energy is released from these linkages.

A liquid fat called oil is obtained from either plants or animals. Made from milk, butter is a solid fat. By taking the milk particles out of butter, ghee is created, which is clarified butter. Due to the high calorie content of all three of these items, it is vital to eat them in moderation. But they may also be a beneficial component of a diet that is balanced.

10. Correct option (C)

There are 9 calories in 1 gram of fat. Compared to proteins and carbs, which both produce 4 calories per gram, this is more than twice as many calories.

Because fat is composed of fatty acids, which are chains of carbon atoms connected by bonds, it has a high calorie content. When the body breaks down the fat, a lot of energy is released from these linkages.

11. Correct option (D)

Heart disease is only one of the health issues caused by excess body fat. The inability of the heart to adequately pump blood is known as heart disease.

Extra body fat raises the risk of cardiovascular disorders like heart attacks, high blood pressure, and stroke and can cause heart issues. Elevated cholesterol levels, insulin resistance, and inflammation are a few risk factors linked to obesity that can put stress on the heart and circulatory system. Maintaining heart health requires maintaining a healthy body weight through a balanced diet and frequent exercise. Chest discomfort, breathing difficulties, and other issues may result from this.

Additionally, high blood pressure, high cholesterol, and diabetes can be brought on by excess body fat. Heart disease risk may also be increased by certain circumstances.

12. Correct option (B)

Based on the solubility of vitamins, vitamins are classified into two types: fat-soluble vitamins and water-soluble vitamins.

Water-Soluble Vitamins: Vitamins that are Water-Soluble: These vitamins may dissolve in water. They consist of the B-vitamins (thiamine, riboflavin, niacin, pantothenic acid, pyridoxine, biotin, folate, and cobalamin) and vitamin C. Water-soluble vitamins are quickly eliminated through urine and are not retained in significant quantities by the body. In order to maintain appropriate levels, frequent consumption is required.

Fat-Soluble Vitamins: Vitamins that are Soluble in Fat: These vitamins are soluble in fat. These vitamins are vitamin A, vitamin D, vitamin E, and vitamin K. Fat-soluble vitamins can be stored in the body's fatty tissues and liver and are absorbed together with dietary lipids. They can collect in the body and, if ingested in excess, may cause toxicity since they are not eliminated as quickly as water-soluble vitamins.

13. Correct option (C)

Calcium is a mineral that is essential for strong bones and teeth. It is also important for muscle function, blood clotting, and nerve transmission.

Calcium is a key element in the formation of bones. In order to guarantee appropriate bone development during infancy and adolescence, while bones are growing and developing, a sufficient intake of calcium is required.

Bone Strength: A sufficient calcium intake lowers the risk of osteoporosis, a disorder characterized by weaker and brittle bones, and helps prevent bone loss.

Teeth Health: Getting enough calcium helps build and maintain healthy tooth enamel, which lowers the risk of dental issues including cavities and sensitivity.
Phosphorus is also important for bones and teeth, but it is not as essential as calcium. Iron is important for red blood cell production, and iodine is important for thyroid function.

14. Correct option (A)

The indigestible component of plant meals is known as roughage or fiber. Dietary fiber, often known as roughage, is a component of plant-based meals such fruits, vegetables, whole grains, and legumes. It is made up of inedible plant cell wall components, namely cellulose, hemicellulose, and lignin.
It gives the faces greater volume and makes it easier for it to pass through the digestive system. This may aid in avoiding constipation.
Constipation cannot be avoided with vitamins, proteins, or lipids. Actually, certain fats might aggravate constipation.

15. Correct option (D)

The primary ingredient in sweat, urine, and faeces is water. It facilitates the removal of waste items from the body and controls body temperature
Sweat: The main component of sweat is water, with trace quantities of minerals and waste materials. Sweating aids in body temperature regulation and toxin removal.
Urine: The kidneys are crucial in the removal of waste and extra water from the circulation. Urine, the ensuing liquid waste, is made up of water and several dissolved materials, including urea, salts, and toxins.
Faeces: Water from the gut is absorbed when food passes through the digestive tract, resulting in the creation of solid waste. Constipation can be avoided and good bowel motions can be maintained with adequate drinking.
Sweat, urine, and faeces do not involve oil or ghee in their creation. Vitamins are necessary nutrients that support the body's ability to operate, but they are not involved in the production of these physiological fluids.
So, the correct option is D) Water

16. Correct option (A) ‘

Excessive water loss from the body results in dehydration. Sweating, nausea, diarrhoea, or a lack of fluid intake are all potential causes. Dizziness, weariness, headaches, and disorientation are just a few of the health issues that dehydration can cause. Dehydration may be lethal in extreme circumstances.
When water vapor in the sky condenses, it precipitates, falling to the earth as rain, snow, sleet, or hail. Water evaporation is the process through which water from the Earth's surface turns into vapor and enters the atmosphere.

17. Correct option (B)

A balanced diet is one that includes all the essential elements in the proper proportions. This indicates that the diet should consist of a variety of foods from all the food categories, including fruits, vegetables, grains, foods high in protein, and dairy.
For the purpose of sustaining excellent health, a balanced diet is crucial. It can aid in the prevention of chronic conditions including heart disease, stroke, type 2 diabetes, and several cancers. Additionally, it may aid in maintaining a healthy weight and enhancing general wellbeing.
A balanced diet is a healthy diet that includes all the necessary nutrients, while protein deficiency is a condition that occurs when the body does not get enough protein.

18. Correct option (B)

Malnutrition is defined as consuming either few or too many of the diet's important nutrients. When there is an imbalance, deficit, or excess of nutrients in the body owing to

insufficient dietary intake, poor nutrient absorption, or elevated nutritional needs, it is referred to as malnutrition.

Different symptoms of malnutrition may appear. Undernutrition is characterized by a lack of or insufficient intake of vital nutrients, which impairs growth and development and causes nutritional shortages. On the other side, overnutrition happens when a person consumes too many calories and other nutrients, which can cause obesity and other health issues.

Hence the correct option is B) Malnutrition

19. Correct option (C)

Kwashiorkor is a severe form of protein-energy malnutrition that develops when the diet is lacking in protein. Young children in impoverished nations that consume diets lacking in protein and other necessary nutrients are frequently affected.

Protein is essential for a number of physiological functions in the body, including growth, tissue repair, immunological function, and others. The body cannot achieve its protein requirements when there is insufficient protein consumption, which leads to the symptoms of kwashiorkor.

Edema (swelling), stunted development, muscular loss, exhaustion, irritability, changes in hair and skin color, and a compromised immune system are some of the signs of kwashiorkor. A well-balanced diet that offers enough protein and other necessary nutrients is used as treatment to alleviate the protein shortage.

20. Correct option (A & B)

A severe case of malnutrition known as marasmus is characterized by a lack of protein, carbs, and other vital nutrients. Children with chronic undernutrition and little access to a healthy food are most likely to experience it.

The development, maintenance, and repair of biological tissues depend on protein, whereas the energy needed for numerous internal processes comes from carbohydrates. Marasmus patients experience severe malnutrition overall, major weight loss, muscle wasting, compromised immune systems, stunted development, and other symptoms as a result of their diet's lack of protein and carbs.

It's critical to understand that marasmus is a complicated illness that involves deficits in more than simply protein and carbs. Additionally, marasmus sufferers lack other crucial components such vitamins and minerals.

21. Correct option (A)

A lack of vitamin B1, also known as thiamine, results in the illness known as beriberi. The neurological system and the metabolism of carbohydrates both depend on the vital vitamin thiamine. It is essential for turning food into energy.

Beriberi frequently develops in areas where polished rice is a staple meal but lacks the essential B vitamins. Wet beriberi and dry beriberi are the two main types. The cardiovascular system is impacted by wet beriberi, which increases the risk of edema and heart failure. Dry beriberi predominantly affects the neurological system and causes symptoms including muscular weakness, lack of coordination, tingling or numbness in the extremities, and trouble walking.

Vitamin B12, vitamin D, and vitamin K are not involved in the metabolism of carbohydrates. Therefore, the only option that is associated with beriberi is vitamin B1. So, the answer is A.

22. Correct option (B)

Scurvy is a condition that arises from a severe deficiency of vitamin C (ascorbic acid). Vitamin C is essential for the synthesis of collagen, a protein that helps in the formation of connective tissues, wound healing, and the health of blood vessels. When there is a lack of

vitamin C in the diet, collagen synthesis is impaired, leading to various symptoms associated with scurvy. The symptoms of scurvy include fatigue, weakness, bleeding gums, and skin problems.

Collagen formation does not include vitamin B1, vitamin D, or vitamin K. Vitamin C is the sole remedy therefore linked to scurvy. Therefore, B is the correct response.

23. Correct option (C)

The absence of adequate red blood cells in the body is known as anemia. Because red blood cells are in charge of transporting oxygen throughout the body, the organism suffers from an oxygen deficit when there are not enough of them.

Red blood cell formation requires vitamin B12, which is necessary. Anemia can develop if the body is unable to manufacture enough red blood cells due to a lack of vitamin B12.

Red blood cell formation does not include vitamin B1, vitamin C, or vitamin K. Therefore, vitamin B12 is the only remedy connected to anemia. The response is C.

24. Correct option (D)

A lack of vitamin D, which is necessary for healthy bone growth and development, can result in rickets. The body's ability to absorb and use calcium and phosphorus is greatly aided by vitamin D.

A vitamin D shortage results in poor mineralization of bones, which causes weaker and fragile bones. Children are frequently affected by rickets, which can result in skeletal abnormalities, stunted development, muscular weakness, and other associated problems.

Vitamins B1, C, and B12 do not contribute to the formation of bones. Therefore, vitamin D is the only supplement that has been linked to rickets. Therefore, D is the correct response.

25. Correct option (D)

A hallmark sign of goiter is enlargement of the thyroid gland. The term "goiter" describes the unnatural expansion of the thyroid gland, which is situated in the neck and generates hormones that control metabolism. Multiple reasons, including iodine shortage, certain drugs, autoimmune diseases, or underlying thyroid issues, might contribute to the growth.

The absence of adequate red blood cells in the body is known as anemia. Red blood cell synthesis is not influenced by the thyroid gland.

Vitamin insufficiency known as beriberi can harm the nerves. The manufacture of vitamin B1, which is lacking in beriberi, does not include the thyroid gland.

A deficiency of vitamin D can cause rickets, a disorder where the bones do not grow normally. The synthesis of vitamin D does not include the thyroid gland.

Therefore, the answer is D) Goiter.

3. Let us understand Plants

Solutions

LEVEL- I

1. Correct option (B)
Buds are small swellings on a plant from which a shoot, leaf, or flower usually develops. They are protected by a covering of modified leaves called scales which tightly enclose the more delicate parts of the bud.

2. Correct option (C)
Rootlets are also called lateral roots and are defined as the small branches of roots that form from the main root and helps the plant for the absorption of water and minerals as they increase surface area. So, the correct option is 'Lateral roots'.

3. Correct option (B)
Fibrous root system is an intricate network of similar sized roots with tiny hair like structures called root hairs. They form a mesh like bundle and does not penetrate deep into the soil. Examples: Maize, Wheat, Grass, etc.

4. Correct option (A)
The petiole is a stalk that connects the blade with the leaf base.

5. Correct option (D)
Stomata which are present on the lower surface of the leaves can also be called as nose of the leaves because through it plants take in Oxygen (O2) and Carbon Dioxide (CO2).

6. Correct option (B)
Ginger is an important and commonly used rhizome. It is a rhizome which is a modified stem of the plant. The rhizome of ginger bears nodes and internodes and scaly leaves are also present.

7. Correct option (C)
Stomata are the tiny, kidney, or bean-shaped pores or openings present in the epidermis of the cell.

8. Correct option (B)
Grasses are monocots and have a fibrous root system. They have a hollow, thin cylindrical stem and the leaves have parallel venation.

9. Correct option (B)
Essentially all the water used by land plants is absorbed from the soil by roots. A root system consists of a complex network of individual roots that vary in age along their length.

10. Correct option (B)
Beetroot forms a tuber that can be elongated, flat-round, round or egg-shaped depending on the variety.

11. Correct option (B)
Plants absorb water and minerals from the soil via their roots. The root hairs on roots increase the surface area of roots to facilitate absorption of water and minerals that are dissolved in the soil water.

12. Correct option (C)
Plants with fibrous roots have parallel venation. Plants that have fibrous roots and parallel venation in leaves include wheat, maize, grass, banana, bamboo, etc.

13. Correct option (B)
The flat green portion of a leaf is called as lamina or leaf blade.

14. Correct option (A)
Veins and veinlets are vascular bundles of lamina. The veins are larger and usually one prominent vein runs from base to apex of lamina and known as midrib. It forms lateral veins and the ultimate branches of veins are called as veinlets.

15. Correct option (B)
Reticulate type of venation is found in dicots hence the root system found in them will be the taproot system.

16. Correct option (C)
Transpiration is the process of removal of excess water from the plants through tiny openings in leaves known as stomata.

17. Correct option (D)
The main function of the leaf is photosynthesis. Photosynthesis is a physio-chemical process in which plants use light energy from the sun to drive the synthesis of organic compounds. Another important process carried out by leaf is transpiration. Transpiration refers to the evaporative loss of water by the stomata present in the leaves.

18. Correct option (D)
Underground stems are non-green stems that may take part in perennation, store food or help in vegetative propagation. Underground stems of potato, ginger, turmeric, Samarkand, Colocasia are modified stem to store food in them. Besides the loss of water vapour in transpiration, exchange of gases takes place through the pores called stomata present on the leaves.

19. Correct option (A)
Stem helps in transportation of food and water to different parts of the plant.

20. Correct option (B)
Midrib is the thickest vein present in the middle of the leaf. It separates the lamina or leaf blade into 2 parts. Smaller veins are observed radiating from the midrib.

21. Correct Matching is: a-Q, b-S, c-T, d-R, e-P

22. Correct Matching is: a-S, b-R, c-T, d-Q, e-P

LEVEL II

1. Correct option (D)
Products like timber, fodder, gums and resin, medicinal plants, fruits and vegetables, etc are procured from the plants and vegetation.

2. Correct option (B)
The plants which have soft and green stems are called herbs. Examples- Cauliflower, wheat, rice, mint etc.

3. Correct option (C)
Herbs are small plants with green and tender stems. The stem has very few branches. These plants have less strength in their stem and can easily be uprooted from the soil. Basil, coriander, Okra and thyme are a few examples of herbs around us.

4. Correct option (B)
Plants which have hard woody stems and branches close to the ground are called shrubs.

5. Correct option (C)
Rose is a shrub since it is a medium sized plant with a hard woody stem and branches out from the base.

6. Correct option (D)
The elongated stem of a tree that supports branches and has a woody outer bark is called the trunk of a tree. Trunk is only present in trees.

7. Correct option (A)
A plant having thin, long and weak stem which cannot stand upright and spreads on the ground is called a creeper.
8. Correct option (B)
Climbers are the plant that climbs on support by developing tendrils are modified stems that help to bind the support.
example-bougainvillaea, Money plant and Grapevine.
9. Correct option (C)
Roots help the plant to absorb water and nutrients from the soil and also provide support to the plant. The plants absorb water from the soil with the help of root hair through osmosis.
10. Correct option (A)
Taproot systems feature a single, thick primary root, called the taproot, with smaller secondary roots growing out from the sides.
11. Correct option (D)
A taproot system comprises one thick, cylindrical main root known as the primary root. Small roots that branch out from the primary root are termed as secondary and tertiary roots. Examples include Turnip, Parsnip, Beetroot, Carrot, Radish.
12. Correct option (A)
Carrots are the modified tap root, for storage. Carrots have a root cap, no nodes, and never directly bear leaves or flowers. Roots serve as food storage organs, absorbing nutrients and moisture, anchoring the plant in the soil, providing physical support for the stem, and anchoring the plant in the soil.
13. Correct option (B)
Aerial roots are roots above the ground. They are almost always adventitious. They are found in diverse plant species, including epiphytes such as orchids, tropical coastal swamp trees such as mangroves, banyan figs.
14. Correct option (D)
The shoot system refers to the parts of the plant growing above ground. This includes the stem, leaves, flowers, Twigs, and buds.
15. Correct option (C)
The stem may not always be typically like what they are expected to be. They are modified to perform different functions. Underground stems of potato, ginger, turmeric, Samarkand, Colocasia are modified to store food in them.
16. Correct option (B)
The fleshy stem of the cactus plant is used to store water. Cacti usually grow in very dry and hot conditions where there is a lack of water resources. Hence, they need to store sufficient water in their stem to meet their needs.
17. Correct option (A)
The spines or sharp needle-like structures seen in cacti are the modified leaves. The actual leaves are modified in order to reduce the surface area which will, in turn, reduce the evaporation of water through stomata.
18. Correct option (C)
In some plants with weak stems, the leaves are modified to form coiled thread-like structures called tendrils. The plants with weak stems can grow horizontally against a support as the tendrils coil around the support.
19. Correct option (C)
Leaves prepare food through the process of photosynthesis. So, they are called the food factories of plants. Photosynthesis is the process by which green plants and some other

organisms use sunlight to synthesize food in the form of glucose using carbon dioxide and water.

20. Correct option (C)

Lamina is the flat and green part of leaf. Lamina is also called leaf blade. Photosynthesis occurs in the lamina of leaf. Lamina contains many linear structures called veins which transport minerals and nutrients to leaf.

21. Correct option (A)

The arrangement of the pattern of veins in a leaf is called leaf venation.

22. Correct option (B)

When the veins in leaves are irregularly distributed to form a channel, it is known as reticulate Venation. Plants with reticulate venation in their leaves are likely to have tap roots.

23. Correct option (C)

Chloroplasts are chlorophyll containing plastids present in leaves of plants that functions as photosynthetic systems to produce sugars.

24. Correct option (A)

Photosynthesis is a process where plant leaves convert carbon dioxide and water into sugars and oxygen is released out in the environment.

25. Correct option (D)

Leaves of plants have small pores known as stomata which aids in gaseous exchange and loss of excess water through transpiration.

4. Nutrition in plants

Solutions

LEVEL – I

1. Correct option (A)

Roots are the plant organs that provide anchorage to the plants and are responsible for the uptake of water and nutrients from the soil & transporting it to the other plant parts. These are specialized and have tiny root hairs that help them to achieve their desired functions.

2. Correct option (B)

Stephen Hales was an English Botanist who first discovered the role of light in plant nutrition. Stephen Hales realized that air could be fixed by plants under the influence of solar light. Priestley showed that plants could regenerate the bad air making it breathable. Ingenhousz demonstrated that the green parts of plants performed the complete purification of air only under the influence of the light.

3. Correct option (A)

Starch is a carbohydrate which is synthesized by plant leaves and is stored as energy reserve by a process known as photosynthesis. To test the presence of starch, Iodine is frequently used. Iodine has the property to bind to Starch molecules and impart a blue-black colour which is an indication of presence of starch.

4. Correct option (A)

Potassium helps regulate the opening and closing of the stomata on leaves, which regulates the exchange of water vapor, oxygen, and carbon dioxide. Potassium also activates enzymes responsible for starch synthesis in plants.

5. Correct option (A)

In plants, minute pores are present on the surface of leaves that serves as the centre for gaseous exchange. The gases diffuse into the intracellular spaces of the leaf through these pores. Gaseous exchange (Air) is necessary for plants for the purpose of photosynthesis and respiration.

6. Correct option (B)

Parasitic plants and fungi have a root-like extension known as Haustorium, that grows into or around another plant to absorb water or nutrients from that plant instead of directly taking it from the soil. Parasitic plants like Dodder and Mistletoe have a haustorium.

7. Correct option (D)

Moist Bread and pickles if left outside, grows a black and white spot on the surface due to spoilage by Fungus, generally by Rhizopus and Mucor species. They generally grow on moist and damp habitat and in warm temperatures.

8. Correct option (A)

Transpiration is the loss of water in the form of water vapour through stomata, cuticles and lenticles. Most of the transpiration occurs through stomata in leaves i.e., 50-90% of total transpiration.

9. Correct option (A)

Urea serves as fertilizers and is a rich source of nitrogen. The urea reacts with the gases present in the soil and is firstly converted into ammonia and then into ammonium. This ammonium is then converted into nitrates and nitrites. Plant leaves turn yellow when there is a deficiency of nitrogen content. Farmers sprinkle urea so that they absorb

nitrogen and fulfill its needs. The leaves soon become green again, therefore increasing the rate of photosynthesis.

10. Correct option (D)

Drocera, Utricularia and Venus fly trap are insectivorous plants that feed on insects as well as their larvae. They generally have modified shoots especially for the purpose of capturing and ingesting the insect.

11. Correct option (B)

Cuscuta also known as Dodder or Amarbel is a genus of parasitic plants which has no chlorophyll, and it absorbs its food and nutrients from its host by wrapping its whole flexible stem around the host plant and absorbing nutrients and stays attached to it until the host dies.

12. Correct option (A)

Parasitic plants and fungi have root-like structures known as Haustorium, that grow into or around another plant and its tissues to absorb water and nutrients from that plant instead of directly taking it from the soil. Parasitic plants like Dodder and Mistletoe have a haustorium as specialised root like appendages projecting out from stem.

13. Correct option (D)

In the process of photosynthesis, Carbon dioxide, CO_2 along with Water, H_2O in presence of solar energy is converted to sugars, specifically Sucrose and O_2 inside the chloroplast present in the leaves of plants. Given below is the reaction.

$6CO_2+6H_2O \rightarrow C_6H_{12}O_6+6O_2$

14. Correct option (C)

Tropical Pitcher plants or Nepenthes is a genus of insectivorous plants that grows in Nitrogen deficient soil and feeds on insects. Nepenthes pitchers produce a lot of sweet nectar to attract insects.

15. Correct option (A)

Drocera is commonly known as Sundew, that are “flypaper” plants that trap prey in sticky hairs on their leaves. They make up one of the largest groups of carnivorous plants. Long tentacles protrude from their leaves, each with a sticky gland at the tip. These droplets look like dew glistening in the sun, thus their name.

16. Correct option (B)

Lenticels are specialized pores present on the surface of stems/ barks and permit the exchange of gases between the environment and the internal tissue spaces of the organs. They permit the entrance of oxygen and simultaneously the output of carbon dioxide and water vapor.

17. Correct option (A)

Stomata are the openings present on the surface leaves that function in exchange of gases in plants continuously. The gases diffuse into the intracellular spaces of the leaf through these pores. Gaseous exchange (Air) is necessary for plants for the purpose of photosynthesis and respiration.

18. Correct option (D)

Carnivorous plants such as Picher plants and Sundews grow in soil which is deficient in Nitrogen. Nitrogen is an essential component for plants to produce proteins and vitamins. Hence, they rely on insects for their nitrogen need.

19. Correct option (D)

Micronutrients are those nutrients that is needed in a small quantity by a body for plant metabolism. There are 7 essential plant nutrient elements defined as micronutrients

[boron (B), zinc (Zn), manganese (Mn), iron (Fe), copper (Cu), molybdenum (Mo), chlorine (Cl)]. They constitute in total less than 1% of the dry weight of most plants.

20. Correct option (C)

In the process of photosynthesis, Carbon dioxide, CO_2 along with Water, H_2O in presence of solar energy is converted to sugars, specifically Sucrose inside the chloroplast present in the leaves of plants and releases O_2 gas outside as by-product. Given below is the reaction.

$6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$

21. Correct Matching is:a-S; b-P; c-T; d-R; e-Q

22. Correct Matching is:a-S; b-R; c-Q; d-P; e-T

LEVEL-II

1. Correct option (A)

The study of different components of diet including carbohydrates, fats, and protein of living organisms is known as science of nutrition.

2. Correct option (C)

Plant leaves have organelles called as chloroplast that synthesis their food requirements by a process known as photosynthesis, where light energy is converted to chemical energy.

3. Correct option (B)

Plants require essential nutrients for proper growth and development. There are in total 17 nutrients that are required by plants. Out of which 6 are macronutrients and others are micronutrients. These are Carbon (C), Hydrogen (H), Oxygen (O), Nitrogen (N), Phosphorus (P), Potassium (K), Sulfur (S), Calcium (Ca), Magnesium (Mg), Boron (B), Chlorine (Cl), Copper (Cu), Iron (Fe), Manganese (Mn), Molybdenum (Mo), Nickel (Ni), and Zinc (Zn).

4. Correct option (A & C)

In the process of photosynthesis, water and carbon dioxide are the precursor material for sugar synthesis by plants and in the process, oxygen is liberated. So, water as well as carbon dioxide cannot be the end product of it.

5. Correct option (D)

Chlorosis is the loss of the normal green coloration of leaves of plants due to loss of chlorophyll. It leads to the yellowing of leaves. This is caused by iron deficiency in lime-rich soils, or due to diseases or lack of light. Other elements that cause chlorosis are Nitrogen, Potassium, Zinc, Manganese, Sulphur etc.

6. Correct option (A)

Dry matter in leafy vegetables and proteins are rich in nitrogen and its molecules such as amines. Thus, Nitrogen is a very crucial mineral that helps in improving both the quantity and quality of that matter.

7. Correct option (D)

In monocots the Guard cell surrounding stomata are dumb-bell shaped, but in case of dicots they are kidney bean shaped.

8. Correct option (A)

Saprophytes are plants/fungi that feed on dead and decaying matter and have heterotrophic mode of nutrition. They lack chloroplast in their cells and thus they aren't green.

9. Correct option (C)

The leaves of the croton plant have chlorophyll, but they appear dark red in color. This is because the higher percentages of other pigments anthocyanins and carotenoid

present in them, hides the green color of the chlorophyll. So, the leaves of the croton plants can also make food by the process of photosynthesis.

10. Correct option (D)

Fungi are ubiquitous that is are found everywhere in this earth, majority of them can grow on the surface of the soil, foods items, in liquid, cloths and leathers, decaying matter etc.

11. Correct option (A)

Plants can synthesize their own food requirement by a process known as photosynthesis, and thus are autotrophic in nature.

12. Correct option (B)

Saprophytes are those organisms that feed and grow on dead and decaying matter. Fungi are such organisms that can grow and derive nutrients from dead leaves and humus in soil. Mushrooms are an example of fungus. Venus's flytrap and the Dodders are insectivorous, Lichen is a symbiotic association between algae and fungi.

13. Correct option (D)

Food that we consume has nutrients and minerals and serves as a mode of nourishment for the body organs for the purpose of energy to carry out metabolism.

14. Correct option (B)

Plant cells have chloroplast, in which stacks of thylakoids form grana and are suspended in matrix known as stroma. Thylakoids have chlorophyll inside which absorbs light and carries out photosynthesis.

15. Correct option (C)

Partial parasitic plants are green as they perform photosynthesis, but they are dependent on other plants for certain nutrients.

16. Correct option (B)

Crows draw their nutrition from both plant and animal sources. The organisms that draw their nutrition from both animals and plants are called omnivores. Hence, the crows are omnivorous organisms.

17. Correct option (A)

Starch is a polymer of sugars present in plants, its presence is detected by iodine, iodine binds to starch turns it into blue-black color.

18. Correct option (B)

Plant hormones are not nutrients but chemicals that in small amounts promote and influence the growth, development, and differentiation of cells and tissues.

19. Correct option (D)

Plant leaves appear green due to chlorophyll present in their chloroplast. Chloroplasts are the sites of photosynthesis in plants and chlorophyll serves as a light absorbing pigment.

20. Correct option (A)

During Photosynthesis, plants use solar energy to convert water and carbon dioxide into sugars and oxygen is released and this sugar is then used as chemical energy during respiration.

21. Correct option (D)

The end product of photosynthesis is Glucose, $C_6H_{12}O_6$ which is formed from water and carbon dioxide in the presence of light.

22. Correct option (B)

The associations of organism where they live together and share shelter and nutrients and are mutually beneficial for each other is known as Symbiosis. Lichen is an example of Symbiosis.

23. Correct option (A)
Except for Mango rest all are heterotrophs, that is they cannot synthesize their own food and must derive their nutrition from other sources.

24. Correct option (D)
Iron and magnesium are essential for the synthesis of chlorophyll in plants. A deficiency of these elements causes chlorotic patches on leaves with the leaves turning yellow.

25. Correct option (B)
They can trap insects because they do not grow in nitrogen-rich soils. Soil lacking nitrogen causes pitcher plants to get nitrogen from the trapped insects.

26. Correct option (A)
The soil of marshy land is deficient in nitrogen. Plants living in marshy areas do not get nitrogen from the soil. To fulfil their nitrogenous need, they need to eat insects.

27. Correct option (B)
These tiny pores present in the leaves are known as stomata. They help in the exchange of gases and give out water vapor through transpiration.

28. Correct option (D)
The associations between organisms where they live together and share shelter and nutrients and are mutually beneficial for each other is known as Symbiosis. Lichen is an example of Symbiosis.

29. Correct option (A)
Rhizobium is a nitrogen fixing bacteria present in the soil that converts atmospheric nitrogen into ammonia which can be taken up by plant roots.

30. Correct option (A)
All green plants have chlorophyll in their cells to prepare their own food by photosynthesis. During the process of photosynthesis, water combines with carbon dioxide to form sugars and oxygen in liberated in the process.

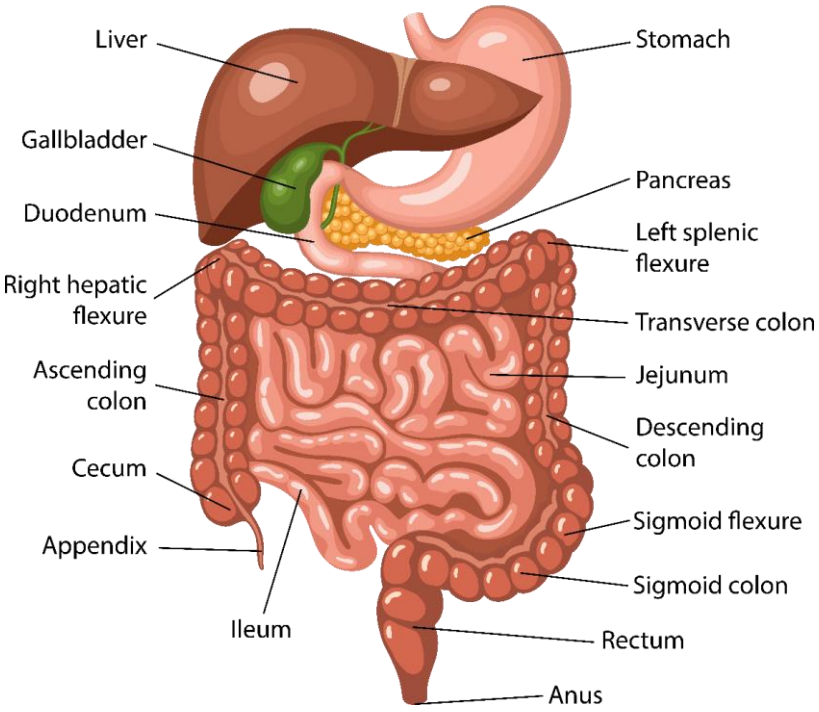
5. NUTRITION IN ANIMALS

Solutions

LEVEL – I

1. Human beings ingest food through the mouth. Digestion begins in the mouth when we chew food. The salivary glands present in the mouth secrete their digestive juice called the saliva. Saliva breaks down the starch present in the food and converts it into sugar. The saliva also keeps the food moist making it easy to swallow. The food that we ingest goes down from the mouth through the oesophagus into the stomach. Stomach receives the partly digested food coming through the food pipe. Digestion of food in the stomach takes one to four hours, depending on the type of food eaten. Digestion of food happens with the help of various secretions such as hydrochloric acid and gastric juice secreted by the walls of the stomach. The semi-digested food, called chyme, leaves the stomach and enters the small intestine. When the food enters the small intestine, it mixes with the bile and the pancreatic juice, secreted into it by the liver and the pancreas, respectively. The small intestine also produces digestive juices, which help to complete the digestion of all the components of food. Here absorption of food takes place in the small intestine. The food that remains undigested and unabsorbed then enters the large intestine. The function of the large intestine is to absorb water and some salts from the undigested and unabsorbed food. The semi-solid food that is left behind is then stored as faeces in the rectum, the last part of the large intestine. The semi-solid faeces is removed from our body from anus.

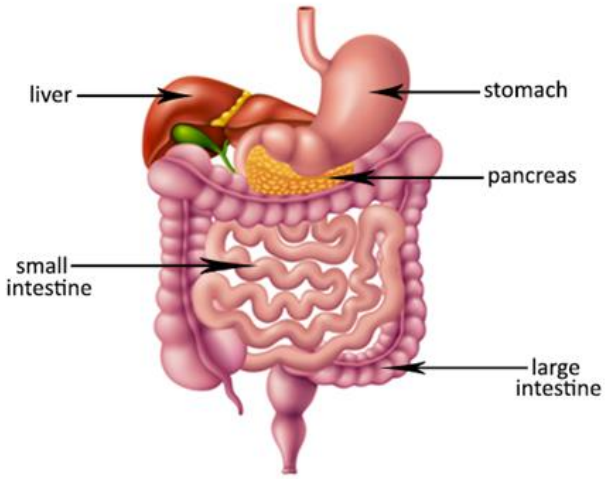
Digestive system



2. Ruminants have a special stomach with four chambers, namely, rumen, reticulum, omasum and abomasum. The swallowed food goes straight into the rumen—the first and the largest of the four chambers in the stomach. The food is partially digested in the rumen. The partially digested food in the rumen is known as cud. The cud is pushed into the reticulum—the second chamber of the stomach and from here it is sent back into the mouth for thorough chewing. The chewed food is swallowed and sent into the omasum—

the third and the smallest chamber of the stomach to absorb excess water from the food. In the omasum, the food is broken down to even smaller pieces. The last chamber—abomasum is called the true stomach. Gastric juices are secreted in the abomasum to help in digestion. Enzymes act upon the food to complete the digestion. The digested food then goes into the small intestine, where absorption of nutrients takes place. A large sac-like structure called caecum present between the small intestine and large intestine also contain bacteria that help in cellulose digestion.

3. 1-e, 2-a, 3-d, 4-b, 5-c
4. False
5. True
6. True
7. True
8. False
9. Ingestion, digestion, absorption, assimilation, and egestion are the five stages of holozoic nutrition.
10. Assimilation
11. The salivary gland secretes digestive juice called saliva which breakdown starch into sugars. It also keeps the food moist and easy to swallow.
12. The antelope swallows grass as the first chamber of its stomach called rumen contains cellulose digesting bacteria which digests the cellulose present in the grass. This is sent to the second chamber and then regurgitated back to the mouth where it is thoroughly chewed.
13. Bile juice
14. I am amoeba. I form a pellet of the undigested food and throw it out of the food vacuole through a temporary opening of the cell membrane.
15. Enzyme
16. Digestion
17. Mouth/Buccal cavity
18. I am any cooked grain such as rice or wheat. I test positive the first time as there is starch present. The second time, I test negative as the saliva breaks down my starch into sugar when I am chewed.
19. The image is villi present on the inner walls of a part of the small intestine. There are millions of finger-like projections of villi which absorb food. They increase the surface area for absorption of food.
20. Pancreas and liver are the two digestive glands. Their secretions are found in the small intestine.



21. Correct option (B)
The whole process starting from ingestion of food by mouth, its breakdown, absorption, and utilization by the body is known as digestion. It's a process by which complex food is taken up by the body.

22. Correct option (B)
Amoeba ingests its food with the help of its false feet or pseudopodia. The food is digested in the food vacuole.

23. Correct option (B)
Human saliva contains salivary amylase which breakdown starch into maltose and dextrin.

24. Correct option (B)
The hydrochloric acid in the gastric juice breaks down the food and the digestive enzymes split up the proteins. The acidic gastric juice also kills bacteria. The mucus covers the stomach wall with a protective coating.

25. Correct option (B)
Digestion and absorption of all biomolecules are completed in the intestine. The small intestine has 3 parts namely- the duodenum (c- shaped), jejunum, and ileum. Complete digestion of food takes place in the ileum. After being digested the food is absorbed by the blood vessels present in the small intestine.

26. Correct option (C)
The millions of finger-like outgrowths on the inner surface of the small intestine are called as Villi. They help in increasing the surface area for the absorption of the digested food.

27. Correct option (C)
Although the small intestine is narrower than the large intestine, it is the longest section of your digestive tube, measuring about 7.5m on average, or three-and-a-half times the length of your body.

28. Correct option (B)
Your body uses amino acids to make proteins. The different types of amino acids and the way they're put together determine the function of each protein.

29. Correct option (C)
Fats are present in the form of large globules in the small intestine. The small intestine receives the secretions from the liver and the pancreas. The bile salts (from the liver) break down the large fat globules into smaller globules so that the pancreatic enzyme lipase can easily act on them.

30. Correct option (D)
Water from the undigested food is absorbed in the large intestine to the body.

31. Correct option (D)
Protein after digestion by proteinases is converted into Amino acids which are absorbed into the bloodstream.

32. Correct option (D)
Digestion of groundnut oil produces fatty acids and glycerol. Groundnut oil is rich in fatty acids.

33. Correct option (D)
The boiled rice and saliva are added to the iodine solution then the colour of iodine will not change because of the digestion of starch present in rice into monomers of sugars.

34. Correct option (B)
Canines are used for ripping and tearing food such as meat.

35. Correct option (D)
Incisors – These are the four teeth which are present in both the upper and lower jaw known as the incisors. These are having the one root. The major function of the incisors is to cut food because they have the sharp edge.

36. Correct option (C)
4 incisors, 2 canines, 4 premolars and 6 molars (Each half of the Jaw has 2,1,2,3 number of teeth's, 2 Incisors, 1 Canine, 2 premolars and 3 molars on each side of the Jaw)

37. Correct option (B)
The root contains blood vessels and nerves, which supply blood and feeling to the whole tooth. This area is known as the "pulp" of the tooth.

38. Correct option (D)
Excessive intake of foods containing sugar causes tooth decay. The germs in the mouth form plaque on the surface of teeth

39. Correct option (C)
Amoebae use pseudopodia, which are arm-like projections on the cell surface, to engulf and ingest food.

40. Correct option (D)
Amoeba is a protist that has pseudopodia or false feet which it uses for locomotion and capturing of food. It has no fixed shape, no fixed mouth and possess no digestive system.

41. Correct option (C)
Peristalsis is a series of wave-like muscle contractions that move food through the digestive tract. It starts in the esophagus where strong wave-like motions of the smooth muscle move balls of swallowed food to the stomach.

42. Correct option (B)
Amoeba is a unicellular organism. It undergoes intracellular digestion that is food is engulfed inside, fused with lysosome and broken down by enzymes inside the cells. The mode of nutrition in Amoeba is holozoic. The process of obtaining food by Amoeba is called phagocytosis.

43. Correct option (B)
Premolars and Molars are the teeth that do most of the chewing because those flat surfaces are perfect for grinding and crushing food until it's ready to be swallowed.

44. Correct option (A)
Holozoic Nutrition is a process in which complex food is taken into a specialist digestive system and broken down into small pieces to be absorbed. This consists of 5 stages, ingestion, digestion, absorption, assimilation, and egestion.

45. Correct Answer: Liver
Bile is a fluid that is made and released by the liver and stored in the gallbladder.

46. Correct Answer: Small intestine
Stomach is connected to Duodenum of small intestine at its lower portion.

LEVEL - II

1. Correct option (C)
Hookworms are parasites that live inside the intestines. There, they feed on blood from the intestinal wall, mature and lay eggs.

2. Correct option (C)
The bees climb onto or into the flower and suck up the nectar with their straw-like mouth and collect it in a little sac called a crop.

3. Correct option (D)
Snakes do not have the right kind of teeth to chew their food so they must eat their catch whole. Their jaw is structured in such a way that it allows the mouth to open wider than their own body to swallow their prey whole.

4. Correct option (A)
Saliva is an extracellular fluid produced and secreted by salivary glands present in the mouth.

5. Correct option (D)
The food vacuole containing the food molecule fuses with lysosome which contains amylase and proteinase enzymes. These enzymes digests carbohydrates and proteins present in the food.

6. Correct option (B)
Hydrochloric acid is the main component of gastric juice and is secreted by the parietal cells or oxyntic cells of the stomach walls in humans.

7. Correct option (D)
The vacuole is pushed deeper into the amoeba by cytoplasmic movements where it fuses with lysosomes - organelles containing digestive enzymes like amylase, cellulase and proteinases which can digest carbohydrates such as Sugars and cellulose and proteins.

8. Correct option (C)
The finger-like outgrowths (Pseudopodia) of Amoeba help ingest food. However, the finger-like outgrowths or villi of human intestines increases the surface area of the walls and thus help to absorb the digested food.

9. Correct option (B)
Cud is the name given to the food of ruminants which is swallowed and partially digested. Cud is a food component that returns from the stomach to the mouth of an animal for a second chew. Rumination is the process by which cud is generated during digestion. Ruminants are other names for cud-chewing animals.

10. Correct option (A)
The large intestine has 3 primary functions: absorbing water and electrolytes, producing and absorbing vitamins, and forming and propelling feces toward the rectum for elimination.

11. Correct option (C)
Digestion is the mechanical and chemical break down of food into small organic fragments.

12. Correct option (D)
In a holozoic mode of nutrition, the organism feeds on complex organic matter. They ingest the complex organic matter, digest, and absorb the nutrients. In animals' mode of nutrition is holozoic because they depend on plants for food, cannot synthesize their own food, thus are heterotrophs, and they swallow their food.

13. Correct option (D)
Lacteals are lymph vessels responsible for absorbing fat in the small intestine. Occurring in the finger-like villi of the ileum, lacteals have a milky appearance and drain into the lymphatic system. Before fat can pass into the lacteal, bile from the liver causes its emulsification into small droplets for action of the enzyme lipase.

14. Correct option (B)
Deciduous teeth are the official term for baby teeth, milk teeth, or primary teeth. Deciduous teeth start developing during the embryonic stage and then commonly begin to come in about 6 months after birth.

15. Correct option (C)
The false feet of Amoeba or the pseudopodia are used to engulf and capture food and also in movement and locomotion.

16. Correct option (B)
The first set of teeth which are lost between the ages of **6 -12 years**, are termed as the milk teeth.

17. Correct option (D)
There are four types of teeth Incisors, Canines, Premolars and Molars.
18. Correct option (A)
In stomach, the medium is acidic with a pH of about 1-2 due to presence of Hydrochloric acid in the Gastric juice. It serves as suitable medium for digestion of proteins in the stomach.
19. Correct option (D)
The removal of faeces through anus is called excretion.
20. Correct option (B)
Premolars and molars pair of teeth differ in structure but are similar in function that is grinding of food.
21. Correct option (A)
Salivary Amylases digest starch into smaller molecules, ultimately yielding maltose, which in turn is cleaved into two glucose molecules by maltase.
22. Correct option (C)
Chyme is a churned semi-solid food that is formed in the stomach of the digestive tract.
23. Correct option (B)
The stomach is a thick-walled muscular bag. Its shape is like a flattened U and it is the widest part of the alimentary canal.
24. Correct option (D)
Although the small intestine is narrower than the large intestine, it is actually the longest section of your digestive tube, measuring about 7.5m on average, or three-and-a-half times the length of your body.
25. Correct option (C)
Proteins are chains of amino acids linked together, once they are broken by proteinases, they yield small units of amino acids.
26. Correct option (D)
Plenty of boiled and cooled water with a pinch of salt and sugar dissolved in it. This is called Oral Rehydration Solution (ORS).
27. Correct option (B)
Euglena has both an Autotropic as well as heterotrophic mode of nutrition. It's a protozoon that can synthesize its own food in presence of sunlight and can capture and consume its prey when deprived of sunlight.
28. Correct option (D)
The tip of the tongue is most sensitive to sweet and salty substances, the side to sour, while the back is most sensitive to bitter taste.
29. Correct option (A)
Villi are finger-like projections in the small intestines that increases surface area for absorption of digested food. They have very thin walls and have network of thin and small blood vessels close to the surface.
30. Correct option (B)
The swallowed food moves downwards in the alimentary canal because of the contraction of muscles in the wall of the food pipe. This movement is under the control of nervous system and is known as peristalsis.

6. Living Organisms - Characteristics & Habitat

Solutions

LEVEL I

1. Correct option (A)

Desert plants and animals are suited to dry conditions and vast temperature differences. Desert plants are adapted to their arid environment in many ways. Small leaves on desert plants help reduce moisture loss during photosynthesis. Small leaves mean less evaporative surface per leaf. In addition, a small leaf in the sun doesn't reach as high a temperature as a large leaf.

2. Correct option (A)

Acacia Nilotic is truly multipurpose nitrogen fixing leguminous tree in India commonly called as babul. Its timber is valued by rural folks, leaves and pod are used as fodder and gum has a number of uses.

3. Correct option (A)

Forests cover 31 percent of the world's land surface, just over 4 billion hectares. (One hectare = 2.47 acres.)

4. Correct option (D)

In Kurnool district we can see a bird called battameka pitta which flies over long distances to come here.

5. Correct option (D)

The main components of a habitat are shelter, water, food, and space. A habitat is said to have a suitable arrangement when it has the correct amount of all of these.

6. Correct option (C)

The methods of reproduction in fishes are varied, but most fishes lay a large number of small eggs, fertilized and scattered outside of the body. The eggs of pelagic fishes usually remain suspended in the open water. Many shore and freshwater fishes lay eggs on the bottom.

7. Correct option (A)

Avian migration is a natural process, whereby different birds fly over distances of hundreds and thousands of kilometres in order to find the best ecological conditions and habitats for feeding, breeding and raising their young.

8. Correct option (B)

Pistoia stratiotes is a free-floating, fast growing, obligate aquatic that can form vast, dense floating mats, covering the entire water surface of lakes and slow-moving rivers

9. Correct option (D)

Rivers are the arteries of the landscape that connect and sustain diverse freshwater wetlands including lakes, inland deltas, swamps and marshes. Safeguarding and restoring healthy, resilient freshwater systems are key to achieving global goals for food, water.

10. Correct option (A)

The lotus is an aquatic plant that grows in shallow fresh water, with flat round leaves that float on the surface and famously beautiful flowers. While the entire plant is edible, the root that grows underwater is particularly popular in Asia, and especially in Japanese cooking.

11. Correct option (B)

Clostridium perfringens, Staphylococcus, and Escherichia coli (E. coli; toxic strain). They inhibit health by triggering disease and promoting aging. Opportunistic bacteria cause no trouble when you are healthy, but have adverse effects upon the intestines when the body is weak.

12. Correct option (C)
Habitats are a very significant part of natural diversity. They offer species a living environment and provide people with essential ecosystem services.

13. Correct option (D)
The main components of a habitat are shelter, water, food, and space. A habitat is said to have a suitable arrangement when it has the correct amount of all of these.

14. Correct option (C)
The common pond skater can be seen 'skating' over the surface of ponds, lakes, ditches and slow-moving rivers. It is predatory, feeding on small insects by detecting vibrations in the water's surface. There are some species of beetles and dragon flies that may at some point in their life cycle live on the surface of water but maximum time they either hover or live on land surfaces.

15. Correct option (A)
The ocean is very, very deep; light can only penetrate so far below the surface of the ocean. As the light energy travels through the water, the molecules in the water scatter and absorb it. At great depths, light is so scattered that there is nothing left to detect.

16. Correct option (D)
Animals move from one place to another in search of food and to protect themselves from their enemies. they also move to escape from the harsh climate. To find shelter and to breed their young ones.

17. Correct option (A)
A cactus is able to survive in the desert due to the following features: (i) It has long roots that go deep inside the soil for absorbing water. (ii) Its leaves are in the form of spines to prevent water loss through transpiration. (iii) Its stem is covered with a thick waxy layer to retain water.

18. Correct option (A)
Grapevines require nutrients for healthy, sustained growth. Without adequate nutrition, vines may have weak growth and reduced yields. An inadequate nutrient supply may be associated with low (<5) or high (>7) soil pH, inadequate irrigation, or competition from weeds or cover crops.

19. Correct option (C)
Mussels are marine or freshwater molluscs that are primarily found in bottom water. They attach themselves to various surfaces such as rocks, shells, or other substrates on the seabed or riverbed. They live in this habitat by filtering water to obtain nutrients.

20. Correct option (A)
An earthworm is a soil-dwelling terrestrial invertebrate that belongs to the phylum Annelida.

21. Correct Matching is: a- Q;b- R;c- S;d- R;e- P
a) – Q) Vallisneria is a genus of freshwater aquatic plant, commonly called eelgrass, tape grass or valise.
b) – R) Terrestrial plants are the plants which grow on land. Mango tree, rose plant and apple tree are terrestrial plants.
c) – S) A mangrove is a shrub or tree that grows mainly in coastal saline or brackish water. Mangroves grow in an equatorial climate, typically along coastlines.
d) – R) cactus is able to survive in the desert due to the following features: (i) It has long roots that go deep inside the soil for absorbing water. (ii) Its leaves are in the form of spines to prevent water loss through transpiration. (iii) Its stem is covered with a thick waxy layer to retain water.
e) – P) Most monkeys live mainly on trees.

LEVEL – II

1. Correct option (C)
All living elements-the birds, animals and plants, forests, fisheries are biotic elements.

2. Correct option (B)
Broadly Biotic components are categorised into two categories: Autotrophs and Heterotrophs or Producers and Consumers. A third category decomposers also exists but it is a part of Heterotrophs. Producers are the plants in the ecosystem, which can generate their own energy requirement through photosynthesis, in the presence of sunlight and chlorophyll. Consumers include herbivores, carnivores, omnivores, Decomposers and Saprophytes. The herbivores are the living organisms that feed on plants.

3. Correct option (A)
The organisms which prepare their own food are called autotrophs or producers. They depend on a process known as photosynthesis for this.

4. Correct option (B)
Photosynthesis is a process by which green plants prepare their own food from carbon dioxide and water in the presence of sunlight and chlorophyll. The reaction involved is as follows: $6CO2(g)+6H2O(l)\rightarrow C6H12O6(s)+6O2(g)$.

5. Correct option (C)
Decomposers are heterotrophic microbes like bacteria and fungi that derive their nutrition by decomposing dead and decaying organisms.

6. Correct option (A)
All animals and fungi are heterotrophs as they can't produce their own food. They all depend on the autotrophs (primary producers) and on other organisms for food. Hence, also known as consumers.

7. Correct option (A)
Jaguars and Tigers are carnivorous mammals that feed mainly on fish (such as piranha), reptiles (like lizards or turtles), birds (turkeys), amphibians (like frogs), as well as small mammals like monkeys and deer.

8. Correct option (C)
As they only eat flesh and meat of other animals for their food, they are classified as carnivores.

9. Correct option (C)
Animals that can consume both plants and other animals for nutrition are called omnivorous animals. Examples: human beings, cockroaches, dogs, bears, cats.

10. Correct option (B)
Humans, bears, crows are referred to as omnivores.

11. Correct option (A)
Plants and animals are interdependent on each other. Plants depend on animals for pollination, and they absorb nutrients from decomposed animal bodies to grow. Animals depend on plants for food, shelter, protection, to get oxygen and so on.

12. Correct option (D)
Earthworms need their skin to breathe. Because their skin is gas permeable. Their skin is moist, and cutaneous respiration is the process of breathing via the skin.

13. Correct option (D)
In roots, stems & leaves the exchange of gases takes place through diffusion. The surface of young stem has stomata, while that of an older stem have lenticel for gaseous exchange. Stomata are mainly present on the surface of leaves apart from the epidermal

surface of young stem for gaseous exchange. In root exchange of gases occurs through root hair.

14. Correct option (A)

The habitats of the plants and animals that live on land are called habitat. Organisms living on the land are called terrestrial animals.

15. Correct option (A)

Terrestrial habitats are ones that are found on land, like forests, grasslands, deserts, shorelines, and wetlands. Terrestrial habitats also include manmade habitats, like farms, towns, and cities, and habitats that are under the earth, like caves and mines.

16. Correct option (A)

Desert sand is a good absorber and good radiator of heat. During the day it absorbs heat quickly and becomes very hot. At night sand lose all the absorbed heat quickly and become cold. Therefore, deserts are cold at night and hot during the day.

17. Correct option (C)

Transpiration is the process through which plants lose water through stomata. Stomata are usually present on the underside of the leaves. It is essentially evaporation of water from the leaves of the plants.

18. Correct option (B)

Succulent plants store water in fleshy leaves, stems or roots. All cacti are succulents, as are such non-cactus desert dwellers as agave, aloe, elephant trees, and many euphorbias. Several other adaptations are essential for the water storing habit to be effective.

19. Correct option (A)

Nocturnal animals are animals that are active during the night and sleep during the day.

20. Correct option (D)

camel have padded feet help camel to walk on hot and slippery sand without sinking.

21. Correct option (B)

High altitudes can cause low oxygen saturation levels or desaturation of an individual's blood. It happens because of low atmospheric pressure at high altitudes. Oxygen saturation levels refer to the extent haemoglobin is bound or saturated to oxygen.

22. Correct option (A)

Herbivores like horses, cows, and rabbits have rows of wide, flat teeth. This helps them in grinding and chewing leaves, grass, and other tough plant materials.

23. Correct option (A)

Many plants in the humid rainforest canopy are pointed, so that rain can run off the tips of the leaves. These “drip tips” keep the leaves dry and free of Mold.

24. Correct option (A)

Habitats whose mode of survival is in water are known as aquatic habitats. The habitats of plants and animals which lives in water are said to live in aquatic habitats. Ocean, ponds, lakes, and rivers are examples of aquatic habitats. If the water is freshwater, then it is called fresh water habitat.

25. Correct option (B).

Large air gaps exist in the parenchyma of aquatic plants to let them float on water. Such a parenchyma type is called aerenchyma. It is a type of tissue with air gaps. It gives hydrophytes buoyancy, allowing them to float to the water's upper surface. These air cavities of aquatic plants allow the plant to be light and buoyant which results in the easy floating of plants in the water.

7. Body Movements
Solutions

LEVEL-I

1. Correct option (D)

When muscles contract, they undergo several simultaneous changes, which is why the correct answer is D) All of these. First off, as muscles contract, the individual muscle fibers within the muscle bundle pull together, shortening the muscle. Movement is made possible by this shortening of the muscle fibers, which also produces force. Second, while contracting, muscles can stiffen up, especially if they are actively resisting an outside force. Increased muscle tone and the activation of motor units cause this stiffness. The compression of muscle fibers and increased tension within the muscle can also make muscles appear thicker during contraction. Therefore, when muscles contract, all of these changes take place, making it a complex physiological process necessary for many bodily functions, such as movement and posture maintenance.

2. Correct option (C)

The bony or cartilaginous structure in vertebrates that is attached to and supports the hind limbs or fins is known as the "pelvic girdle," as indicated in option C. In vertebrates, the pelvic girdle is an essential part of the skeletal system, and it differs in structure depending on the species. The pelvic girdle, which forms the hip bones in terrestrial vertebrates like mammals, including humans, is crucial for supporting the body's weight, serving as points of attachment for leg muscles, and allowing hind limb movement. Similar functions are performed by the pelvic girdle in aquatic vertebrates like fish, which stabilize and support the pelvic fins that facilitate swimming and movement. While options A (backbone), B (ribs), and D (vertebrae) are also important parts of the vertebrate skeletal system, they do not specifically refer to the structure supporting the hind limbs or fins, making option C the correct choice for this description.

3. Correct option (B)

The delicate, fleshy structures that surround and are found beneath the skin are known as muscles. They are in charge of posture, movement, and heat generation. Muscle fiber bundles form the structure of muscles. Long, thin cells that can contract and relax are called muscle fibers. A muscle that is attached to a bone shortens and pulls on that bone when it contracts. This is how muscles cause bones to move. The body cannot function without muscles. They enable us to move, keep our posture, and generate heat. We could not survive without our muscles.

Tendons, which connect muscles to bones through fibrous cords, are strong. Bands of fibrous tissue called ligaments attach bones to one another.

4. Correct option (D)

The answer, "All the above," accurately reflects that almost all our body movements depend on the combined function of muscles, bones, and joints. Joints enable the range of motion and flexibility needed for various types of movements, while bones provide structural support and act as attachment points for muscles. Muscles contract and relax to generate the force required for movement. Our ability to perform a wide range of activities, from the fundamental activities of daily life to more difficult physical tasks, is due to the complex coordination between our muscles, bones, and joints. Our body's musculoskeletal system is a marvel of functional integration, allowing us to effectively interact with and navigate our environment. Whether it's the contraction of our leg muscles during walking, the flexibility of our shoulder joints for reaching, or the stability provided by our spinal bones.

5. Correct option (C)

Cartilage, as identified in option C, is a type of connective tissue that can be considered a bone-like structure, but it possesses flexibility. Contrary to rigid bones, cartilage is malleable and offers some flexibility and shock absorption in various body parts. It acts as a structural element where it forms the joint surfaces, allowing for painless movement in places like the nose, ears, and ends of long bones. Although cartilage is not as hard as bone, the body still needs it to provide support and protection. Although neither of these structures is regarded as flexible like cartilage, ligaments (option A) are fibrous tissues that connect bones to other bones, and tendons (option B) connect muscles to bones. Therefore, option C, "Cartilage," which is a flexible tissue that resembles bone, is the appropriate response.

6. Correct option (A)

The skull is also called the "cranium," as correctly identified in option A. The bony cranium surrounds the brain in a protective framework, giving it vital support and preventing damage from the outside world. In addition to housing the brain, it also houses other sensory organs like the eyes and ears. It is made up of several bones that are tightly fused together to form a sturdy, protective enclosure for the brain. The largest portion of the brain is referred to as the "cerebrum" (option C), while the "brain" (option B) refers to the entire organ housed within the cranium. Therefore, when referring to the skeletal framework of the head, option A, "Cranium," is the appropriate response.

7. Correct option (B)

Muscle contraction causes the muscle fibers to shorten, causing the muscle to thicken and enlarge. The reason for this is that the muscle fibers are shortening as a result of pulling against one another. The muscle becomes thicker and more bowed the more the muscle fibers shorten.

The opposite of contraction is relaxation. The muscle fibers lengthen as a muscle relaxes, making the muscle thinner and less puffed up.

So, the muscle bulges due to contraction. Option A, relaxation, is incorrect because the muscle does not bulge when it relaxes. Option C, both A & B, is incorrect because only contraction causes the muscle to bulge. Option D, none of these, is incorrect because the muscle does bulge due to contraction.

8. Correct option (D)

Cartilage can be found in the vertebral column, nose, and ears. Hyaline cartilage, elastic cartilage, and fibrocartilage are the three types of cartilage found in the ear. The eardrum and the joints between the bones of the middle ear contain hyaline cartilage. The outer ear is composed of elastic cartilage. The joint between the jawbone and the skull contains fibrocartilage.

Both hyaline and elastic cartilage can be found in the nose. The wall that separates the nose into two nostrils, known as the septum, is made of hyaline cartilage. The pinna, which is the outer portion of the ear, and the nostrils both contain elastic cartilage.

Vertebrae, stacked one on top of the other, make up the vertebral column. The joints between the vertebrae are cushioned by intervertebral discs, which are made up of fibrocartilage.

9. Correct option (C)

These vertebrae are a part of the spine's vertebral column, which acts as the body's main structural support system. The cervical (neck), thoracic (upper back), lumbar (lower back), sacral (pelvic), and coccygeal (tailbone) regions are among the regions that make up the vertebral column. In a typical adult spine, which has 26 vertebrae in all, there are 7 cervical, 12 thoracic, 5 lumbar, 5 fused sacral, and 4 fused coccygeal. These vertebrae offer

adaptability and stability, enabling a variety of movements and safeguarding the spinal cord, an essential component of the central nervous system.

10. Correct option (D)

The human ribcage plays a crucial role in protecting the heart and lungs, so the correct response is D) Both A & B. The sternum and ribs make up the bony structure known as the ribcage (breastbone). It creates a cage-like shield around the chest region. The ribs surround and protect the heart, a vital organ in charge of pumping blood throughout the body, as well as the lungs, which are essential for breathing. The ribcage is crucial in protecting the heart and lungs because any blow to the chest can potentially damage these important organs.

11. Correct option (B)

Vertebrae, not vertebrates, are the name for the tiny bones that run along the back and support the spinal cord. A group of animals known as vertebrates have a backbone or vertebral column made up of these individual vertebrae. In vertebrate animals, including humans, the spinal column is made up of bony segments called vertebrae. The delicate spinal cord, a vital component of the nervous system that transmits signals from the brain to the rest of the body, is shielded by these vertebrae, which also support the body's structural integrity. Therefore, the individual bones that make up the spinal column or backbone in vertebrate animals are referred to as "vertebrae."

12. Correct option (C)

Muscles and bones are the two main parts of our body that help in body movements.

The tissues that contract and relax to move the bones are called muscles. Tendons connect them to the bones. The human body has over 600 muscles, which account for about 40% of our body weight.

The rigid, brittle structures that give the body support and structure are called bones. Additionally, they shield the internal organs. The adult human skeleton has 206 bones.

In order to move, the muscles and bones must cooperate. The muscles supply the force to move the bones while the bones serve as a framework for the muscles to attach to.

Therefore, bones and muscles both play a role in how the body moves. They work together to produce the movements that we need to do everyday activities, such as walking, running, and eating.

13. Correct option (C)

The correct answer is C) Both A & B. Through a combination of expansion (stretching) and contraction, muscles are what move bones. When a muscle contracts, it shortens and pulls on the bone to which it is attached, moving that bone as a result. It is this contraction that produces the power required for various bodily movements. On the other hand, when a muscle contracts or shortens, it prevents the bone from moving or returning to its original position. As a result, muscles' expansion (relaxation) and contraction are both involved in the process of moving bones, making Option C the best option.

14. Correct option (A)

Ligaments are strong, fibrous connective tissues that provide a unique means of securing bones to one another. They are essential for stabilizing joints because they link one bone to another. Ligaments are crucial for preserving the skeletal system's structural integrity and ensuring that joints work properly. In contrast to muscles, which contract and produce movement, ligaments support joints and restrict excessive motion, thereby reducing the risk of dislocations and other joint injuries. Therefore, ligaments are in charge of securing bones so that joints remain stable and allow for coordinated, controlled movement.

The lower leg's calf bones are those. The fusion of bones is not their responsibility. Tendons connect muscles to bones. Collagen also makes up tendon tissue. The force of a muscle contraction is transferred to the bones by the tendons, which causes the bones to move.

15. Correct option (C)

The answer, "C) Pivotal," is accurate because the pivotal joint is specifically found in the neck region. This kind of joint permits rotational movement about a central axis, which is necessary for the neck's flexibility and mobility. It gives us the ability to turn our heads side to side and look in various directions. For many daily tasks and activities, such as driving, reading, talking, and taking in our surroundings, this rotational ability is essential. The pivotal joint in the neck is a crucial component of the musculoskeletal system of the human body because it allows for smooth and controlled rotation, enabling a variety of head movements.

16. Correct option (B)

Fixated joints, also referred to as sutures or immovable joints, are those where the bones are closely connected and do not permit any significant movement. These fixed joints or sutures hold the cranium's bones together in the human skull. This immobility is crucial for preserving the head's overall stability and shape while safeguarding the sensitive brain inside the skull. The joints in the vertebral column (Option A), the neck (Option C), and the shoulder (Option D) all allow for varying degrees of movement, whereas the joints in the skull (Option B), which are fixed, remain rigid to protect the brain and facial structures.

17. Correct option (A)

The correct answer is A) Forward push by pressing the ground. Snails have a distinctive and slow form of locomotion. They cover the underside of their bodies with a unique muscular structure called a foot. A snail moves by tensing the muscles in its foot, which causes it to push forward against the ground. Mucus produced by the foot reduces friction, allowing the snail to move forward more easily. Snails move slowly and steadily because of this forward and rhythmic pushing against the surface and the lubricating impact of the mucus. They don't move by jerking or forming food (Option B) (Option C). Therefore, snail movement is primarily accomplished by pressing their muscular foot into the ground.

18. Correct option (C)

A hinge joint is indeed found in the elbow and arms. Similar to how a hinged door opens and closes, this kind of joint allows movement primarily in a single plane. The elbow joint allows the forearm to bend and straighten in relation to the upper arm in the case of the elbow. It is essential for many daily activities, such as bending the elbow to lift objects or extending the arm to reach for something, that this hinge-like structure maintains stability while enabling controlled and relatively simple movement in a single direction. The hinge joint is thus appropriately named given its resemblance to a hinge and its function in allowing the elbow and arms to move.

19. Correct option (A)

Tendons are the connective tissues that join muscles to bones. They act as the connecting element that enables the force produced by contracting muscles to be transferred to the bones, resulting in movement at the joints. Tendons are strong and adaptable, built to withstand the stress and tension brought on by muscle contraction. Contrarily, cartilage (Option C) is a different type of connective tissue that covers the ends of bones at joints and provides cushioning and support, whereas ligaments (Option B) connect bone to bone and stabilize joints. So, Option A is the best choice because tendons are specifically responsible for joining muscles to bones.

20. Correct option (D)

The average infant's backbone contains 33 vertebrae. Considering that infants have extra, unfused vertebrae, this number is different from the 26 vertebrae that make up the adult spine. In contrast to adults, infants have 5 unfused sacral vertebrae and 4 unfused coccygeal vertebrae in their spines, which have 7 cervical vertebrae, 12 thoracic vertebrae, and 5 lumbar vertebrae. These sacral and coccygeal vertebrae gradually fuse together as a baby grows, giving rise to the 26 vertebrae that make up an adult spine. As an infant's body develops into adulthood, the number of vertebrae changes, which is a typical stage of development.

21. Correct Matching is: a-Q, b-R, c-P, d-S, e-T

a) Pivotal joint - Q (Neck)

One axis can be rotated around by a pivot joint. The head can rotate from side to side thanks to the neck's pivot joint.

b) Hinge joint - R (Elbow)

A hinge joint only permits movement in one direction, such as when a door is opened and closed. Arm flexion and extension are made possible by the elbow's function as a hinge joint.

c) Ball and socket joint - P (Shoulder)

Movement in all directions is possible with a ball and socket joint. As a ball and socket joint, the shoulder enables the arm to move in a variety of directions and lift, reach, and rotate.

d) Fixed joint - S (Cranium)

No movement is possible at a fixed joint. The bones of the skull are connected by a fixed joint called the cranium.

e) Ligament - T (Connects one bone to another)

A ligament is a tissue band that joins two bones together. Ligaments aid in stabilizing joints and limit excessive joint movement.

22. Correct Matching is: a-R, b-P, c-Q, d-T, e-S.

a) longest bone - R (Femur)

The longest bone in the human body, the femur, is the thighbone. It joins the hip to the knee and is situated in the upper leg. The femur is in charge of bearing the body's weight and enabling movement. The strongest bone in the body is the femur. It is composed of a dense, fracture-resistant network of bone tissue.

b) smallest bone - P (Stirrup)

The tiniest bone in the human body is called the stirrup. It is about 3 millimeters long, located in the middle ear. From the eardrum to the inner ear, sound vibrations are transmitted by the stirrup. The stapes is another name for the stirrup. It is one of the middle ear's three ossicles. The malleus and incus are the other two ossicles.

c) largest bird - Q (Ostrich)

The largest bird in the world is the ostrich. It can weigh up to 150 kilograms and reach heights of 2.7 meters. The only remaining species of the genus Struthio is the ostrich, which is native to Africa. The ostrich is a bird without wings, but it can run quickly. It is capable of travelling at up to 70 kph.

d) smallest bird - T (Hummingbird)

The world's tiniest bird is the hummingbird. It weighs less than 2 grams and can reach a length of up to 10 centimeters. Native to the Americas, hummingbirds can be found in a range of environments, including forests, mountains, and deserts. The hummingbird can hover in midair and flies extremely quickly. It is also an excellent pollinator and aids in the transfer of pollen from one flower to another.

e) Backbone - S (Vertebrae)

The row of bones that runs down the back is known as the backbone. It consists of 33 vertebrae stacked on top of one another. In order to connect the vertebrae, muscles and ligaments are used. The backbone supports the body's weight and permits movement. Additionally, it safeguards the spinal cord, a network of nerves that passes through the middle of the back.

LEVEL-II

1. Correct Option (B)

The organisms made up of a single cell are known as "unicellular," as correctly identified in option B. In unicellular organisms, there is only one cell, which is responsible for all vital life processes. Although these organisms' structures are frequently straightforward, they can be highly specialized and environment-adaptive. Protozoa, certain varieties of algae, bacteria, and some fungi are examples of unicellular organisms. As mentioned in option A, multicellular organisms, on the other hand, consist of numerous cells arranged into tissues, organs, and organ systems, allowing for increased complexity and specialization. In this situation, Option C, "Both A and B," is incorrect because it implies that the response could be either unicellular or multicellular, which it is not.

2. Correct Option (A)

Amoeba is an example of a unicellular organism, as indicated in option A. In unicellular organisms, a single cell performs all necessary life processes on its own. The protist amoeba is a well-known instance of a unicellular eukaryote. Amoeba has a straightforward physical structure, but it exhibits a variety of complex behaviors, including movement, feeding, and reproduction, all of which are carried out inside the boundaries of its single cell. Unicellular organisms like Amoeba are vital parts of the microbial world, playing a variety of roles in ecosystems and academic research. They can be found in a variety of environments, including freshwater, soil, and even inside the human body.

3. Correct Option (B)

The term used to describe organisms made up of more than one cell is "multicellular," as identified in option B. The characteristic of multicellular organisms is the presence of numerous cells arranged into tissues, organs, and organ systems. These cells collaborate to carry out specific tasks and add to the organism's overall structure and function. The evolution of multicellularity is a significant evolutionary change that has increased biological complexity and diversity and given rise to a diverse range of organisms, including fungi, animals, plants, and more. As mentioned in option A, unicellular organisms, on the other hand, are made up of just one cell that is capable of performing all vital life processes on its own. In this situation, Option C, "Both A and B," is incorrect because it implies that the response could be either unicellular or multicellular, which it is not.

4. Correct Option (C)

The correct answer is C. Plants and Animals. Animals and plants are both examples of multicellular organisms, which are made up of many cells that cooperate to form intricate structures. Plants can perform vital processes like photosynthesis and reproduction because their cells are arranged into tissues, organs (such as leaves, stems, and roots), and organ systems. On the other hand, animals display a variety of multicellular body plans as well as specialized tissues and organs for activities like digestion, respiration, and locomotion. Option C is the best option because it correctly identifies the two main groups of multicellular organisms in the biological world: plants and animals. These multicellular characteristics set plants and animals apart from unicellular organisms like Amoeba, as found in options A and B.

5. Correct Option (A)

The term used to describe the movement of animals from one place to another is "locomotion," as correctly identified in option A. Animals move in a variety of ways, or locomotion, including by walking, running, flying, swimming, or crawling. It is an essential part of an animal's life that allows them to forage and mate as well as explore their surroundings and avoid predators. Animals' ability to move around is a crucial adaptation that allows them to live and thrive in their respective ecosystems.

An organism's mobility is its capacity to move. It is a more inclusive term than locomotion and can refer to the movement of tissues, cells, and organs. Space between two points is measured as distance. It is not a substitute for movement.

6. Correct Option (D)

The statement is correct, as animals move for a variety of reasons, encompassing all the options provided. First off, animals frequently move in order to meet their basic needs for food and shelter. In order to locate suitable resources and habitats, they might need to explore various locations. Second, moving is an essential defense mechanism against enemies and predators. Animals have the ability to escape danger and improve their chances of surviving by hiding or running. Last but not least, movement is necessary for locating potential partners for reproduction. Many animals must find mates in order to breed, and this frequently entails moving to particular locations where potential partners can be found. In light of these various factors, the response "D. All the above" adequately encapsulates how important movement is to animals' survival strategies.

7. Correct Option (A)

The organ system that supports the body is the skeletal system, as mentioned in option A. The skeletal system is the body's intricate network of bones and cartilage, and it plays many important roles. Its main function is to support the body structurally, keeping the body in shape and safeguarding important organs like the heart, brain, and lungs. In addition, the skeletal system aids in movement by acting as points of attachment for muscles, allowing us to walk, run, and engage in a variety of activities. Additionally, it is essential for making blood cells and storing minerals, especially calcium and phosphorus. The skeletal system serves as the system that supports and underpins our physical form because it is crucial for the integrity and functionality of the human body as a whole.

8. Correct Option (B)

The animals with a backbone are known as vertebrates, as mentioned in option B. Vertebrates are a diverse group of creatures with internal skeletons that are fully developed and include a vertebral column or backbone. They differ structurally from invertebrates, which have no vertebral column. Vertebrates, which include numerous well-known animals like fish, mammals, birds, reptiles, and amphibians, occupy a variety of ecological niches on Earth. Their vertebral column is a distinguishing feature of this group because it provides stability, defense for the spinal cord, and the framework for the rest of the body.

9. Correct Option (A)

Animals without a backbone are known as invertebrates, as indicated in option A. Unlike vertebrates, which have a distinctive bony or cartilaginous vertebral column, invertebrates are a large and diverse group of animals. From simple organisms like sponges and jellyfish to more complex animals like insects, mollusks, and arachnids, this group contains a diverse range of species. Invertebrates play crucial roles in ecosystems all over the world by displaying a remarkable range of adaptations and occupying different ecological niches. Invertebrates are an intriguing and significant component of the animal kingdom because they rely on a variety of structural and support mechanisms as opposed to vertebrates, which have a clearly defined internal skeleton.

10. Correct Option (B)

The earthworm is indeed an example of an invertebrate, as stated in option B. Animals without a vertebral column or backbone are known as invertebrates. Earthworms are creatures with long, segmented bodies that are members of the phylum Annelida. They are a sizable portion of the enormous group of invertebrates that comprise most animal species on Earth. Vertebrates, on the other hand, are animals with a well-developed internal skeleton that includes a vertebral column or backbone, as was mentioned in option A. Because they lack this internal skeletal system, earthworms are classified as invertebrates. Earthworms should therefore be classified as invertebrates rather than vertebrates.

11. Correct Option (B)

An exoskeleton is a hard covering that protects the body of an invertebrate. It is constructed of the tolerant, flexible material chitin. Exoskeletons are present in most invertebrates. Insects, spiders, crustaceans, and mollusks are included in this. The exoskeleton supports the muscles, protects against predators, and helps to keep the body in shape. Invertebrates periodically shed their exoskeleton as they develop. This is known as moulting. The invertebrate can grow by moulting instead of shattering its exoskeleton. Invertebrates that lack an exoskeleton include worms and jellyfish. The soft bodies of these invertebrates are shielded by additional structures, such as a shell or mucus.

12. Correct Option (C)

Earthworms have tiny hair-like structures called "bristles," as mentioned in option C. The ventral surface of the earthworm's body is covered in these bristles, also referred to as setae. Chitin, the same substance used to create insect exoskeletons, is used to create the setae. They can be extended and retracted, and they are arranged in pairs along the earthworm's body. They are essential for the earthworm's movement and burrowing processes. The worm is held firmly in place by the bristles, which also help to grip and propel it through the tunnels it digs underground. Earthworms can move in a coordinated, wave-like motion by alternately contracting and relaxing the muscles associated with their bristles. This enables them to efficiently burrow through the soil, feed on organic matter, and carry out other crucial tasks.

13. Correct Option (B)

Snails have two pairs of tentacles, as mentioned in option B. On their head, snails have sensory organs called tentacles. Depending on the species, the eyes, which are carried by the upper pair of tentacles, can be straightforward or more intricate. The lower pair of tentacles primarily uses its senses of taste and touch to detect and explore its surroundings. These tentacles are essential for helping snails move about, find food, and detect potential dangers.

14. Correct Option (A)

The body of a cockroach is indeed divided into three parts: the head, thorax, and abdomen. The sensory organs, such as the eyes, antennae, and mouthparts, are housed in the head. The legs and wings (if any) are attached to the thorax, which is the middle section of the body. The digestive, reproductive, and respiratory systems are located in the abdominal region, which is the back of the body. Cockroaches and many other insects have anatomical structures that are distinguished by this three-part division. As a result, "Thorax," option A, is the right response.

The larynx, which is a component of the respiratory system, is absent from the cockroach's body. Tentacles are sensory organs, and cockroaches lack them on their bodies.

15. Correct Option (B)

Two sets of wings are attached to the breastbone of birds. The primary feathers are on the first pair of wings, and the secondary feathers are on the second pair of wings. The larger

and more significant of the two pairs, the primary feathers are used for flight. The secondary feathers, which are smaller, aid in balancing the bird while it is in flight. Feathers, which are joined to the wing's bones, make up a bird's wings. Lift and drag are produced by the arrangement of the feathers. Lift is the force that pushes the bird up into the air, and drag is the force that opposes the motion of the bird through the air.

The wings of a bird are an essential adaptation that allows them to fly. They are also used for balance and steering. Birds do not have two pairs of legs. They have one pair of legs, which are located on the underside of the body. The legs are used for walking, running, and perching.

16. Correct Option (A)

All vertebrates have a hard inner structure or framework supporting their body called a "skeleton," as indicated in option A. The skeleton is an integral part of the vertebrate body plan and performs a number of crucial tasks. It gives the body its shape and structural support, safeguarding important organs like the heart, lungs, and brain. The skeleton also provides a system of levers and serves as points for muscles to attach, which helps with movement. Additionally, it contributes to the development of red blood cells and the storage of minerals like calcium and phosphorus. While the precise makeup and organization of the skeleton may differ between different species of vertebrates, it is a characteristic that unites all of them.

The skeleton contains the flexible tissue known as cartilage. Although it is stronger than muscle, it is not as hard as bone. Cartilage is found in the joints, where it helps to cushion the bones and allow for movement. It is also found in the nose, ears, and larynx. Tendons are tough cords of tissue that connect muscles to bones. They help to transmit the force of muscle contraction to the bones, which allows for movement.

17. Correct Option (C)

Option C, "Fish," indicates that a fish has fins and a tail, which are specialized adaptations that allow it to swim effectively inside water. Fish can change direction, maintain balance, and modify their depth thanks to the stability and control that fins give them in the water. The primary fish propelling mechanism is the tail, also known as the caudal fin. It generates the thrust required to move forward in an aquatic environment through its back-and-forth motion. These characteristics make fish well-suited for life in aquatic habitats, whether it be in oceans, rivers, lakes, or ponds. They also help them navigate through water with ease and precision.

18. Correct Option (C)

A fish's skeleton is covered in muscles. Fish have an internal framework of bones or cartilage called an endoskeleton, which serves as structural support for the fish's body. A fish's ability to move and move around is greatly influenced by its muscles, which are attached to its endoskeleton. Fish can swim, change directions, and perform a variety of other movements by flexing their skeletal components by contracting and relaxing their muscles. Therefore, a fish's ability to move and control itself depends on the muscles that cover its endoskeleton.

19. Correct Option (D)

Snakes have rectangular scales that line the underside of their body, making option D, "Snake," the correct answer. The snake's locomotion is greatly aided by these scales, also known as ventral scales or belly scales, which are distinct from the scales on the snake's back. These belly scales are arranged in a smooth, overlapping pattern that gives snakes good traction and grip on a variety of surfaces. They are well adapted to their serpentine way of life thanks to this particular scale arrangement, which enables them to slither across the ground, climb trees, and navigate their surroundings.

20. Correct Option (B)

Birds have a spindle-shaped (streamlined) body which reduces air resistance. Air resistance is the force that opposes the motion of an object through the air. A body is said to be streamlined if its surface is smooth and curved, as this lowers air resistance. A bird's spindle-shaped body reduces air resistance in two ways. First, the body's curved, smooth surface aids in reducing turbulence in the air flow surrounding it. The air moving erratically in turbulence causes drag. Second, the spindle-shaped body helps to minimize drag by directing air flow over the body. Having less air resistance allows birds to fly more effectively. This is due to the fact that flying a bird uses less energy. Birds with spindle-shaped bodies can therefore fly for longer stretches of time and cover greater distances.

21. Correct Option (A)

The upward movement during the flapping of wings is called the "upstroke," as mentioned in option A. A continuous cycle of wing movements is used by birds to produce lift and thrust during flight. As it gets ready for the upcoming downstroke, the bird's wing moves upward during the upstroke, usually against the resistance of the air. In order to maintain aerodynamic stability and control during flight, the upstroke is essential. The wing then moves downward during the downstroke, producing the main force required for lift and propulsion. Together, these back-and-forth up-and-down movements produce the necessary airflow patterns for birds to fly effectively and navigate through the air.

22. Correct Option (B)

Birds' strong chest muscles play a key role in the up-and-down movement of their wings during flight. In order to produce the required force for flapping their wings, birds have highly developed pectoral muscles. The lift and thrust needed by birds to take flight and maintain altitude are produced by the up-and-down motion of their wings. One of the key adaptations that allows birds to achieve powered flight and perform a variety of aerial manoeuvres is the development of these specialised chest muscles. Many animals, in contrast to birds, lack these specialized wing muscles, which prevents them from being able to fly in the same way that birds can.

23. Correct Option (B)

When a bird lands, perches on a branch, or walks on the ground, its tail feathers are crucial in helping it stay balanced. The tail, which is made up of all these feathers, stabilizes the bird's body while it engages in various activities. The tail feathers of a bird aid in directing and stabilizing its flight. But even when perching, landing, or walking on the ground, the tail feathers balance the animal by changing their position and angle as necessary. The ability to maintain balance is necessary for the bird to maintain its posture, avoid falls, and perform tasks with accuracy, making the tail feathers an important part of a bird's anatomy.

24. Correct Option (C)

Penguins, particularly those found in Antarctica, have wings that have been modified into flipper-like structures that are ideal for swimming. Although penguins are flightless birds and therefore cannot fly in the conventional sense, their wings have developed to function as strong underwater flippers. They can swim through the icy waters of the Antarctic region with great agility and efficiency thanks to their modified wings, pursuing and catching prey like fish and krill. Because they spend a large portion of their lives in the ocean and depend on their specialized wings to survive in their aquatic environment, penguins must be able to swim in order to survive.

25. Correct Option (D)

The answer is D. 206 and 600. An adult human body has 206 bones and over 600 muscles. The bones form the skeleton, which provides support and structure for the body. The muscles are attached to the bones and allow for movement.

Everybody's body contains a slightly different number of bones. Due to birth defects or injuries, some people may have more or fewer bones. However, an adult human body has 206 bones on average.

The human body's muscles are also variable. Because of their genetic makeup or level of physical activity, some people may have more or fewer muscles. However, an adult human body has more than 600 muscles on average.

Together, the bones and muscles support and enable movement in the body. The muscles attach to the bones as a framework, and they contract and relax to move the bones. The bones also protect the organs and tissues of the body.

26. Correct Option (B)

The right response is option B, "22," because the human skull is made up of 22 bones. The cranium, which houses and shields the brain, and the facial bones, which make up the structure of the face, are two of these 22 bones. The brain and other sensory organs, like the eyes and ears, are protected by the skull, which also serves as the framework for the face's features. The skull is a crucial component of the skeletal system.

The portion of the skull that houses the brain is known as the cranium. There are 8 bones in it:

Occipital bone, Sphenoid bone, Ethmoid bone, Frontal bone, Parietal bone (2), Temporal bone (2), and Parietal bone (2)

The bones that make up the face are the facial bones. They are made up of 14 bones:

Maxilla (2), Mandible, Zygomatic bone (2), Nasal bone (2), Lacrimal bone (2), Palatine bone (2), Inferior nasal conchae (2), Vomer, Inferior turbinate

27. Correct Option (C)

Option C, "33," is the right response because the vertebral column is made up of 33 ring-shaped bones called vertebrae. The spinal column, which runs down the back and gives the spinal cord vital support and security, is made up of these vertebrae stacked on top of one another. The cervical (7 vertebrae), thoracic (12 vertebrae), lumbar (5 vertebrae), sacral (5 fused vertebrae), and coccygeal regions of the vertebral column are each comprised of five vertebrae (4 fused vertebrae). Together, these vertebrae function as an essential component of the human skeletal system because they safeguard the spinal cord while also enabling flexibility and movement.

28. Correct Option (D)

The ribcage is a cage-like structure of bones that protects the heart and lungs. The 24 ribs that make up this structure are joined to the sternum (breastbone) in the front and the spine in the back. Although the ribcage is strong enough to shield the heart and lungs from harm, it is also flexible enough to allow for breathing.

Blood is pumped throughout the body by the muscular heart. It is situated directly behind the sternum in the middle of the chest. We can breathe thanks to the lungs, which are two spongy organs. They are situated on either side of the heart, and the mediastinum and heart are what divide them (the space between the lungs).

By absorbing shock and avoiding crushing them, the ribcage shields the heart and lungs from harm. It also helps to keep the lungs expanded, which is necessary for breathing.

29. Correct Option (C)

The limbs' bones are supported by girdles. Girdles are bodily parts of people that attach the limbs to the axial skeleton (the central part of the skeleton that includes the skull, spine, and ribcage). The pectoral girdle, which joins the upper limbs (arms) to the axial skeleton, and the pelvic girdle, which joins the lower limbs (legs) to the axial skeleton, are the two main girdles in the human body. These girdles are essential for supporting and stabilizing the limb bones, which enables a variety of movements and functions like walking, running,

and lifting. Therefore, girdles are vital for the proper functioning and mobility of the limb bones, making option A, "Limbs," the correct answer.

30. Correct Option (C)

The longest bone in the human body, which is located in the thigh, is called the "femur." The main bone of the upper leg is the femur, also known as the thigh bone, which runs from the hip joint to the knee joint. Its length and strength are crucial for giving the lower body support, stability, and mobility. Its length and strength make it essential for maintaining stability and balance while moving through various physical activities like walking and running.

The flexible tissue that cushions joints is cartilage. Tendons are strong tissue cords that attach muscles to bones. Tendons and cartilage are not bones.

8. Adaptation of Animals to Climate
Solutions

LEVEL-I

1. Correct option: (A) Very hot and wet

The climate found in Kerala is categorized as "very hot and wet". Accordingly, Kerala has high temperatures and humidity all year round. The area has high rainfall throughout the monsoon season, which adds to its profusion of wildlife and beautiful foliage. The distinctive natural beauty and picturesque landscapes of Kerala are mostly due to its warm and humid climate, which also makes it a great location for a variety of plant and animal species to thrive.

2. Correct option: (A) In the afternoon

When the sun rises, the temperature reaches its peak for the day. The Earth's surface starts to warm when the sun rises, which raises temperatures throughout the morning and early afternoon.

Although the maximum temperature is normally measured around midday when the sun is at its highest point in the sky, this heating impact peaks in the late afternoon. After then, when the sun sets and the Earth radiates heat away, the temperatures begin to steadily fall. Thus, the hottest time of the day often occurs in the morning, around noon, before it begins to drop off in the evening.

Early in the day, when the sun is low in the sky and its rays are less direct, the temperature is at its lowest of the day. The air above the Earth's surface has cooled down throughout the course of the previous night.

3. Correct option: (B) Sand cools down faster

Nights in deserts are much cooler primarily because of the low heat-retaining capacity of sand. During the day, the desert sand absorbs a significant amount of heat from the sun and becomes very hot. However, as the sun sets and the desert loses its direct exposure to sunlight, the sand cools down rapidly due to its low heat retention properties.

This leads to a substantial drop in temperature, making desert nights considerably cooler compared to the scorching daytime temperatures. Option (b) correctly identifies this phenomenon as the reason for cooler nights in deserts.

The other choices are incorrect. In deserts, the humidity does not rise significantly at night. Although wind does blow in deserts, it does not blow hard enough to considerably chill the air. Deserts have no nighttime precipitation.

4. Correct option: (A) Climate

Climate is the annual average of a location's long-term average temperature and precipitation. A region's climate is made up of the regular weather factors that prevail there for a prolonged period of time, often 30 years or more, including temperature, precipitation, humidity, and other atmospheric components.

It gives us insight into the typical weather patterns and trends in a region, enabling us to distinguish between various climatic zones like tropical, temperate, dry, etc. Climate is the proper identification of this phrase in Option (a).

The other choices are incorrect. Climate is the long-term average of a region's weather conditions. The state of the atmosphere at a specific location and time is called the weather. The technique of anticipating future weather conditions is called weather forecasting.

5. Correct option: (D) All of these

Temperature, humidity, and rainfall are a few examples of weather components. The term "weather" describes the transient atmospheric conditions and changes that take place

from day to day or even hour to hour. These factors are very important in influencing the local weather at any given time. Other weather factors may include cloud cover, visibility, wind speed, air pressure, cloud cover, and atmospheric conditions like fog or frost. Option (d) is correct as it includes all the mentioned elements of weather.

6. Correct option: (D) All of these

Forecasting the weather has many uses and benefits in daily life. By giving people early warnings and enabling them to take the required steps to preserve lives and property, it assists in preparing for adverse weather conditions, such as storms, hurricanes, or excessive rains.

It helps with daily activity planning since it allows people to plan outside events, travel, and other activities in accordance with the weather prediction. Weather forecasting is also essential for organizing agricultural activities since it allows farmers to arrange irrigation, planting, and harvesting operations depending on the expected weather. The right response is option (d), which contains all of the advantages of weather forecasting that have been highlighted.

7. Correct option: (D) All the above

The humidity of a place refers to the amount of water vapor present in the air. It is an indicator of the amount of moisture in the air and has a big impact on how comfortable or uncomfortable people feel in different weather situations. Low humidity can result in dryness, which can have an adverse effect on the health of the skin, the respiratory system, and some materials. High humidity can make the air seem heavy and stuffy.

Precipitation, or falling water from the sky, includes both rainfall and snowfall. Humidity and precipitation are not the same. Precipitation is the amount of water that falls from the sky, whereas humidity is the amount of water vapor in the air.

8. Correct option: (C) Polar region

Penguins have evolved to exist in the polar regions, especially in Antarctica and its surroundings. They are equipped to endure in severely cold and icy conditions. Penguins have various adaptations that allow them to survive in these extreme temperatures, including a coat of insulating blubber, thick feathers that trap air for further insulation, and unique glands to remove extra salt from their bodies. Their distinctive body composition and streamlined form also help them swim and dive more effectively in frigid waters. The polar area is where penguins are evolved to reside, hence option (c) is the right response.

Equatorial areas are hot and humid, whereas rainforests are warm and moist. Deserts are hot and dry. Because they are not suited to the heat or the humidity, penguins would not be able to live in these environments.

9. Correct option: (B) Rajasthan

The state of Rajasthan, which lies in the northwest of India, is renowned for its hot and dry weather. It frequently encounters summertime temperatures far beyond 40^0 Celsius (104 degrees Fahrenheit), which are considered to be severe. The area is known for its extremely low rainfall and dry and semi-arid climate. Rajasthan is among the hottest and driest places in India due to its climate. The Thar Desert, the largest desert in India, is located in Rajasthan.

The other choices are incorrect. The climates of Kashmir are chilly and mountainous, Kerala is hot and humid, and the North East has a humid, subtropical environment.

10. Correct option: (B) They cannot swim

Penguins are well-suited to live in the water and have excellent swimming abilities. Being excellent swimmers, they move through the water with tremendous agility using their wings, which have changed into flippers. Penguins spend a large amount of their lives in

the water where they forage for fish and other marine life. They are widely recognized for their swimming prowess.
The other statements are all correct. Penguins have webbed feet, streamlined bodies, and do snuggle together. Penguins keep warm by huddling together, and their webbed feet and sleek bodies enable them to swim and dive.

11. Correct option: (C) Desert regions

The adaptive traits listed in the query, such as being nocturnal, having big, cushioned feet, sweating less, and having water-saving systems, are traits frequently observed in animals living in desert environments. Animals living in deserts may survive in their environment despite the intense heat and lack of water because to these characteristics.
Numerous desert animals are nocturnal, meaning they are most active at night when it is cooler and less water evaporates. Desert animals are able to avoid sinking into the sand because to their wide, cushioned feet, which help them disperse their weight over a broader area. Because sweating causes water loss and desert animals must preserve water, they do not sweat as much. Some desert animals have unique water-saving strategies to aid them. For instance, camels can create water when necessary thanks to the fat that is stored in their hump.

12. Correct option: (A) Climate

The average weather pattern taken over a long period, such as 25 years or more, is known as "climate." The term "climate" refers to the normal atmospheric conditions, such as temperature, precipitation, humidity, wind patterns, and other meteorological factors, that are long-term fixtures in a certain area. It helps us comprehend the typical weather patterns encountered in many parts of the world and gives us a larger perspective on the long-term weather trends in a certain place.
Weather refers to the atmosphere's present condition at a specific location and time. The process through which a species or creature adapts to its environment over time is change. Humidity refers to how much water vapor is present in the atmosphere.

13. Correct option: (A) Polar region

The coldest region on Earth is the polar region, specifically the areas near the North Pole and the South Pole. Due to their proximity to the Earth's poles and the prolonged darkness of the winter, these areas suffer very frigid temperatures all year long. The main polar regions recognized for their arctic climates, ice, and snow are Antarctica and the Arctic, both close to the poles.
On the other hand, tropical locations are close to the equator and receive the most direct sunshine, which causes the temperature to rise as well. The temperate zones, which lie in between the tropical and polar regions, are more temperate than any of them. Near the oceans are the coastal zones. Coastal places often have milder temperatures than inland areas because the seas serve to regulate local temperatures there.

14. Correct option: (A) Fatty tissue

The hump of a camel is a reservoir of fatty tissue, not water or milk. The capacity of camels to store fat in their humps, which they may use as an energy reserve during periods of food scarcity, is widely documented. Contrary to common opinion, camels are extremely adept at conserving water and may go for extended periods without drinking. Instead of acting as a water storage organ, the hump. During its travels through dry and desert areas, the camel receives vital nutrition and energy from the fat tissue in the hump. The camel's bodily fluids, not its hump, are where it stores water. The kidneys of the camel are excellent at retaining water. The camel's mammary glands, not its hump, are where milk is produced.

15. Correct option: (A) Pressure

The appropriate response is (a) Pressure. Usually, a daily weather report does not include pressure. In order to convey information about the present or predicted weather conditions, the daily weather report often concentrates on factors like Temperature (b), Rainfall (c), and Humidity (d). Understanding the current atmospheric conditions and making appropriate plans for your everyday activities depends on these factors.

Although crucial for analyzing weather patterns and formulating forecasts, pressure is not often covered in a brief and uncomplicated daily weather report intended for the general audience. Instead, to evaluate and forecast more complicated weather systems, meteorologists frequently study and analyses pressure values and trends.

16. Correct option: (C) Meteorological department

The department that prepares the weather report is the meteorological department. It is in charge of tracking, evaluating, and predicting weather patterns. To provide precise and fast weather reports and predictions, the meteorological department gathers data from numerous weather stations, weather satellites, and other sources. These reports are essential for many industries that rely on weather data, including public safety, aviation, agriculture, and many others.

A nation's telecommunications infrastructure, including phone lines, cell towers, and internet cables, is the responsibility of the telecom department. A nation's electrical system, which distributes electricity to residences and businesses, is under the control of the electrical department. Buildings, bridges, and other public works projects are within the purview of the civil department.

17. Correct option: (D) Moisture

The amount of humidity in the atmosphere is a measure of the moisture content in the air. In order to evaluate weather conditions and determine how comfortable it is for people to be outside, it is crucial to know the air's humidity level. Our sense of temperature can be affected by the air feeling damp and high humidity levels. On the other side, low humidity can result in dry air, which can have an adverse effect on the skin, respiratory health, and some materials.

To determine how hot or cold something is, use the term "heat." A substance's temperature is a measure for the amount of heat it contains. Materials comprised of atoms or molecules are known as chemicals.

18. Correct option: (B) Rain gauge

The instrument used to measure rainfall is called a rain gauge. It is a simple meteorological instrument made to gather and measure the total amount of precipitation—mostly rainfall—over a predetermined time frame. Although there are many other types of rain gauges, the most popular one is a cylindrical container with markings showing the quantity of rainfall in inches or millimeters. Rain comes into the gauge, gathers in the container, and the height of the water gathered reflects the quantity of precipitation that fell during the specified time period.

An instrument used to gauge atmospheric pressure is a barometer. An instrument used to measure temperature is called a thermometer. A maximum-minimum thermometer is a device that measures the temperatures at their highest and lowest points throughout a specific time period.

19. Correct option: (C) Maximum – minimum thermometer

The maximum and minimum temperatures of a day are recorded using a maximum-minimum thermometer. A 24-hour day is typically the time span for which this sort of thermometer is intended to monitor and record the peak and lowest temperatures. Two distinct scales—one for monitoring the highest temperature and the other for the

minimum temperature—and a U-shaped glass tube make up the device. The liquid expands as the temperature rises and travels up the tube, revealing the highest temperature achieved. The liquid, on the other hand, contracts as the temperature decreases and flows down the tube, indicating the minimum temperature.
Both a Celsius and a Fahrenheit thermometer are used to monitor temperature, however neither one records the peak and lowest temperatures for a certain time period. Although a hydrogen thermometer may measure extremely low temperatures, it is uncommon to use one to note the high and low temperatures for a given day.

20. Correct option: (B) Sun

All the changes taking place in the atmosphere are primarily caused by the (b) Sun. The fundamental energy source for the Earth's weather and climate systems is the Sun. The uneven heating of the Earth's surface caused by solar radiation leads to the creation of air masses, wind patterns, and weather events. The water cycle, which produces clouds and precipitation through the evaporation and condensation of water vapor, is likewise fueled by the energy of the Sun.
The atmosphere's changes are not significantly influenced by the Moon, Earth, or stars. The tides are impacted by the Moon's gravity, although the atmosphere is not significantly affected. There are minor changes in the atmosphere brought on by the Earth's rotation, although these changes are minimal. The atmosphere is not much impacted by the stars at all.

21. Correct option: (A) Polar regions

The places where the sun does not rise for six months and does not set for the other six months are the (a) polar regions. These areas are close to the poles of the Earth, notably the North and South Poles. The tilt of the Earth's axis causes continuous darkness for roughly six months during the polar winter and continuous sunshine for about six months during the polar summer. The Earth's axis is tilted in relation to its orbit around the Sun, which affects the length of the day at various latitudes throughout the year.
This phenomenon is not present in the equatorial, northern, or southern areas. Even though the amount of sunshine varies throughout the year, the sun rises and sets in these areas every day.

22. Correct option: (D) - 37 °C

The polar regions are the coldest regions on Earth. The average temperature in the polar regions in winter can be as low as -37°C. Due to the tilt of the Earth's axis, the sun's rays during the winter months strike the polar areas at a relatively oblique angle. This implies that the Earth's surface is not warmed by the sun's beams for as long.
Snow and ice in the polar regions reflect the sun's rays back into space. The surface of the Earth gets further cooled as a result. Since land surrounds the polar regions, it does not have the same cooling impact as oceans. Oceans absorb heat, preventing extremes in temperature.

23. Correct option: (D) Weather

he short-term variation that occurs in the atmosphere is called "weather." The daily variations in temperature, humidity, precipitation, wind speed, and cloud cover that can occur over short periods of time are referred to as weather. It is the state of the atmosphere at a specific location and time, offering details about the current circumstances.
Contrarily, climate refers to the typical weather patterns for a certain place over an extended period of time. Usually, the length of a climate measurement is 30 years. The quantity of water vapor in the air is known as humidity. What determines how hot or cold anything is its temperature.

24. Correct option: (B) altitude or height above sea-level

The climate at a place depends on various factors, but one of the most significant factors is altitude or height above sea-level. The atmospheric pressure drops as altitude rises, resulting in thinner air. Weather patterns and temperature are directly impacted by this alteration in atmospheric circumstances. In general, temperatures are cooler at greater altitudes, which explains why mountains and high plateaus typically have colder climates than low-lying locations.

While elements such as air pressure, wind speed, and temperature affect climate in crucial ways, the height or elevation above sea level frequently has a decisive impact on the general climatic features of a given region.

25. Correct option: (D) Polar bear

The animal that possesses blubber under its skin is the (d) Polar bear. Blubber is a thick layer of fat found under the skin of certain marine mammals, such as whales, seals, and polar bears. It serves as an insulating layer, helping these animals to stay warm in cold environments, particularly in the frigid waters of the polar regions. Polar bears, in particular, rely on their blubber to maintain body heat and survive in the Arctic's harsh and icy conditions.

There is no blubber under the skin of red-eyed frogs, elephants, or birds. In tropical jungles, there exist tiny amphibians called red-eyed frogs. In savannas and woodlands, elephants are huge creatures. Feathered creatures with wings are birds. Due to their climatic adaptations, none of these creatures require blubber to keep warm.

LEVEL-II

1. Correct option: (C) Toucan

A big, long beak is one of the distinguishing characteristics of toucans, which is known about. In addition to its amazing size, its beak has rich coloring that is frequently decorated with vivid colors. In addition to other functions, a toucan's beak enables it to reach fruits and seeds that are higher up on branches than it is physically able to reach. It also participates in communication, territorial displays, and assists in mate attraction.

Pigeons, penguins, and kiwis don't have very lengthy beaks. In order to dig for food, kiwis have long, narrow beaks. The small, stubby beaks of penguins are employed for capturing fish. Pigeons utilize their short, pointed beaks to devour seeds and other small foods.

2. Correct option: (D) Catch its prey

A sharp beak is a specialized weapon used by eagles to effectively grab and eat their prey. Eagles are raptors with exceptional hunting skills, and their beaks are built to shred and tear apart their prey's flesh. They can grip and hold onto their catch tightly thanks to their sharp, hooked beak.

Additionally, it helps the eagle swiftly break off bite-sized chunks of meat, making it simpler for it to devour. While eagles occasionally migrate, they are not known for hiding from or fleeing from predators, and their beaks are mostly specialized for their carnivorous diet and hunting ability.

They are apex predators, meaning they are at the top of the food chain and lack natural predators, therefore option A is to flee from them. Because they are huge birds, they are not particularly adept at hiding from predators, thus choose option B. Eagles migrate, option C, although their beak does not aid them in this endeavor.

3. Correct option: (D) All of these

Numerous elements, such as average temperature, soil type, and yearly precipitation, have an impact on the variety of species in varied ecosystems. Each of these elements has a significant impact on the habitat's environmental characteristics, which in turn impacts the kinds of species that may survive there. Different species have distinct adaptations

that enable them to endure and procreate in certain environmental circumstances. For instance, certain species could do better in hot, dry habitats with particular soil types, whereas others might do better in cold, damp conditions with various soil traits. The sum of these elements leads to the total biodiversity seen in different environments across the world.

4. Correct option: (B) Orchids

The term "epiphyte" refers to plants that grow on top of other plants (often trees) without taking nutrients from those plants. The most popular epiphytes are orchids. With the aid of specialized adaptations, they may adhere to tree trunks or branches and draw water and nutrients from the surrounding trash, the rain, and the air.

The popularity of orchids in horticulture and decorative plant collections can be attributed to their well-known for having an array of exquisite blossoms. The well-known tree species fir and pine do not develop as epiphytes on other plants, despite the fact that they are not epiphytes.

Both fir and pine are tree species, not epiphytes. They are rooted to the ground and develop roots as they grow in the soil.

5. Correct option: (B) Hibernation

Hibernation is the term for when animals sleep or go into a state of reduced activity in the winter to preserve energy and endure severe weather. Many animals, especially tiny mammals like bears, groundhogs, and bats, hibernate as a seasonal adaptation to withstand lower temperatures and scarcer food sources. The animal may use its stored energy stores to survive until the warmer months return since during hibernation its metabolic rate and body temperature drop dramatically to fit the environment. Torpor enables animals to endure the harsh winter months when food is in short supply.

Different methods that animals adapt to their surroundings include camouflage, movement, and mimicry. With the use of camouflage, an animal can conceal itself from predators by blending in with its environment. An animal can move about its surroundings through locomotion. Animals can mimic other organisms to look like them, frequently for trickery.

6. Correct option: (D) All of these

In addition to India, Malaysia is located in Southeast Asia, Brazil is in South America, Kenya is in East Africa, the Republic of Congo is in Central Africa, Indonesia is in Southeast Asia, Nigeria is in West Africa, and Uganda is in East Africa. Warm temperatures, high humidity levels, and year-round rainfall in these areas often provide the perfect environment for the luxuriant and diversified ecosystems seen in tropical rainforests. Numerous plant and animal species found nowhere else on Earth may be found in these jungles, which are renowned for their extraordinary biodiversity

7. Correct option: (D) - 37 °C

The polar regions are the coldest regions on Earth. The average temperature in the polar regions in winter can be as low as -37°C. Due to the tilt of the Earth's axis, the sun's rays during the winter months strike the polar areas at a relatively oblique angle. This implies that the Earth's surface is not warmed by the sun's beams for as long.

Snow and ice in the polar regions reflect the sun's rays back into space. The surface of the Earth gets further cooled as a result. Since land surrounds the polar regions, it does not have the same cooling impact as oceans. Oceans absorb heat, preventing extremes in temperature.

8. Correct option: (B) Camouflage

A turtle may camouflage itself by blending in with its surroundings using the colors and patterns on its shell. The ability to blend in with their environment allows animals of

many different kinds to use camouflage, which makes it more difficult for predators to identify them and for prey to detect their presence.
In the case of turtles, the patterns and hues of their shells frequently match the forest floor or other aspects of their natural habitat, giving them some kind of defense against possible dangers. By lowering their probability of being seen by predators or prey, this adaptation boosts their odds of surviving.
Some creatures go into hibernation during the winter months, which is a condition of inactivity and low metabolic rate. Animals travel from one location to another during migration, frequently in pursuit of food or a better environment. Behavior is how an animal acts or responds to its surroundings.

9. Correct option: (C) Average in weather pattern in a few years

Climate is the long-term average of the weather in a certain area over a lengthy period of time, usually spanning decades or even centuries. It considers variables including temperature, humidity, precipitation, wind patterns, and other long-term atmospheric conditions. For a variety of uses, including agriculture, urban planning, and environmental research, climate provides important information that helps us understand the normal or predicted weather patterns in a particular place.
On the other hand, weather is the momentary condition of the atmosphere. From day to day, or even from hour to hour, it might alter. Many of the same elements that affect climate also affect weather, although it is also impacted by more regional causes like thunderstorms and hurricanes.

10. Correct option: (A) A white fur, fat below skin, keen sense of smell

Polar bears can survive in extremely cold settings because to a number of adaptations, including:
Their fur is really translucent, giving it the appearance of being white, which helps them conceal themselves when hunting and blend in with their snowy surroundings.
Polar bears have a thick layer of blubber under their skin that serves as insulation and aids in the preservation of body heat in the chilly Arctic conditions.
Polar bears have a keen sense of smell that enables them to locate seals and other prey at a distance. This skill, which enables them to find food even when it is concealed beneath the ice, is essential to their survival in their frigid habitat.
The alternative choices are incorrect. Large eyes, thin skin, and a long tail would all be problems for bears in the cold. Large eyes would make it difficult for bears to see in the snow. The bears' huge, white paws would aid in swimming but not in keeping them warm.

11. Correct option: (B) Swimming in water

Frogs and ducks have webbed feet as a form of adaptation to help them swim effectively in water. Their feet have larger surface area thanks to the webbing between their toes, enabling them to push against the water with more power. This helps them swim and move through watery surroundings. For ducks and aquatic frogs who spend a lot of time in the water, this adaption is extremely advantageous.
The other options are not correct. Frogs and ducks do not need their webbed feet to hold branches tightly, fly in the air, or grab and eat insects.

12. Correct option: (C) Adaptation

An organism's ability to grow and live in a given climate or habitat is referred to as adaptation. These characteristics are the product of natural selection and evolutionary processes, in which qualities that offer an organism a competitive edge in a particular environment is favored and handed down through generations. In order to properly obtain food, shelter, and reproduce within their ecological niche, organisms undergo adaptations to deal with elements including temperature, precipitation, humidity, and other

environmental circumstances. Therefore, an organism's fitness and success in a given environment are greatly influenced by its adaptations.
The alternative choices are inaccurate. Climate variables including temperature, humidity, and wind are referred to as the short-term weather. Climate is the collection of long-term, typical weather patterns. None of these words discuss the specific traits that an organism possesses that make it suitable to a certain environment.

13. Correct option: (A) 15 º – 40 º C

The temperature in the tropics ranges from 59 º to 104 º Fahrenheit, or 15 º to 40 ºC. Near the equator, tropical zones receive warm to hot temperatures all year long. Due to these areas' close closeness to the equator, where they receive constant, bright sunshine, the temperature there stays relatively high. Depending on the location, altitude, and other local conditions, the precise temperature within this range might change, but tropical locations often have warm climates with high humidity levels that sustain a wide variety of ecosystems and lush flora.

14. Correct option: (B) Arboreal fi burrow

Burrows and arboreal habitats are two separate categories of animal habitats. Burrowing animals dwell underground in tunnels or shelters, whereas arboreal creatures live in trees. Some creatures, like squirrels, may dwell in both burrows and trees, therefore these two habitat types are not mutually exclusive. The pair Arboreal fi Burrow is incorrect since the two phrases are not equivalent.
The other choices are the right ones. Animals classified as terrestrial or aquatic or amphibian live on land, in water, or both.

15. Correct option: (C) Long grasping tail

All the other choices (A, B, and D) are traits that are typically associated with creatures that are evolved to surviving in cold conditions, such as the Arctic or polar areas, which is why C) is the odd one out.
A) Thick layer of fat under the skin: Polar bears and seals use this adaption, known as blubber, to help them insulate themselves.
B) White fur: As a kind of camouflage, several animals have evolved white fur to assist them blend in with their environment.
D) Animals with broad, big feet and keen claws, such as polar bears, frequently exhibit this adaptation, which makes it easier for them to move around and grasp frozen surfaces and to successfully acquire prey.
Animals living in cold climates are not often thought to have long, gripping tails. For animals that dwell in trees or deep woods and depend on holding on to branches for support or movement, a long gripping tail may be more appropriate. It is not a trait frequently observed in creatures that have evolved to cold regions.

16. Correct option: (B) They cannot swim

Penguins have exceptional swimming skills, making them well-suited to life in the water. They are adept swimmers and use their wings, which have become flippers, to navigate through the water with great dexterity. Penguins spend a significant portion of their lives foraging for fish and other aquatic creatures in the ocean. They are also known for having excellent swimming skills.
All of the other claims are true. Penguins do cuddle up together and have webbed feet and streamlined bodies. Penguins can swim and dive thanks to their webbed feet and streamlined bodies, and they keep warm by huddling together.

17. Correct option: (C) To blend in and hide

The ability to change color allows chameleons to blend in with their environment. This enables them to ambush prey and hide from predators. They can adjust their skin color to

match the colors of nearby objects, such as leaves or branches, which helps them remain unnoticed. Chromatophores are peculiar skin cells that are present in chameleons. These cells carry pigment in a variety of colors. These cells are really growing or shrinking as a chameleon changes color. This alters how much pigment is exposed, which in turn alters the skin color of the chameleon.
The other options are not correct. Chameleons do not change color to use less energy, to change their temperature, or to help them hibernate.

18. Correct option: (A) Highest day temperature and lowest night temperature of the day

The highest recorded temperature during the day, often in the afternoon, is referenced by the maximum temperature that is shown in the weather broadcast. The highest temperature that day was represented by this.
The minimum temperature listed in the weather broadcast is the lowest reading made overnight or early in the morning, usually just before daybreak. The lowest temperature that day was represented by this.
The alternate choices are incorrect. Usually, the weather bulletin does not include the month's highest daytime or nighttime temperatures. Additionally, it is not customary to include the temperature recorded at noon and at midnight because these are not necessarily the highest and lowest temperatures of the day. The average highest and lowest temperatures of the day and night are also not often shown since they are less useful than the actual highest and lowest temperatures.

19. Correct option: (C) Red-eyed frog and beard ape

Red-eyed frogs and beard apes are both found in tropical rainforests, which are hot and humid environments. The rainforests provide these animals with the moisture and shade they need to survive.
All around the world, tropical rainforests are home to red-eyed frogs. They can see in the dark thanks to their vivid red iris. Additionally, sticky pads on the foot of red-eyed frogs enable them to climb trees.
The Western Ghats of India are the home of the bearded ape. They go by the name "lion-tailed macaques." A long, thick mane of hair in bearded apes serves as sun protection. Additionally, they have a long tail that aids in balance while climbing trees.
Elephants in Africa and polar bears can only be found in cold regions; hence they cannot live in hot and humid settings. The tundra and desert, which are both cold and dry habitats, are where you may find camels and reindeer.

20. Correct option: (C) Average weather pattern of many years

Climate is the long-term, typically spanning decades or even centuries, average of the weather in a particular place. Temperature, humidity, precipitation, wind patterns, and other long-term climatic conditions are all factors that are taken into account. Climate offers crucial information that helps us comprehend the typical or anticipated weather patterns in a certain location for a range of tasks, including agriculture, urban planning, and environmental research.
Contrarily, weather refers to the current state of the atmosphere. It might change from day to day, or even from hour to hour. The same factors that influence climate also have an impact on weather, albeit it is also influenced by more localized factors like thunderstorms and hurricanes.

21. Correct option: (D) All of these

Tropical rainforests are characterized by hot and humid climates, enormous numbers and varieties of animals, and competition for food and shelter.

The environment in tropical rainforests is often hot and humid, with average temperatures of 15 ° – 40 ° C. Additionally, there is a lot of humidity, with typical percentages ranging from 70 to 90%
Huge variety and quantity of creatures: The tropical rainforests are home to a huge diversity and number of animals. Tropical rainforests are home to about half of all species on Earth, or an estimated 10 million different kinds of plants and animals.
Competition for food and shelter is a result of the high levels of biodiversity seen in tropical rainforests. These creatures may evolve specialized adaptations as a result of this intensive competition.

22. Correct option: (A) They cannot swim

Penguins can swim quite effectively, which makes them perfect for a life in the sea. They are skilled swimmers and can navigate through the water with incredible agility using their wings, which have evolved into flippers. Penguins forage on fish and other aquatic animals in the ocean for a sizable portion of their life. They are also renowned for their prodigious swimming abilities.
The other assertions are all accurate. Penguins have webbed feet, streamlined bodies, and they do snuggle up together. Penguins' webbed feet and streamlined bodies enable them to swim and dive, and they keep warm by huddling together.

23. Correct option: (C) Minimum and maximum thermometer

A unique kind of thermometer known as a minimum-maximum thermometer is used to track the daily minimum and maximum temperatures. There are two bulbs in it, one loaded with mercury and the other with alcohol. The minimum temperature is recorded using an alcohol bulb, while the highest temperature is recorded using a mercury bulb.
The temperature of a person or a material is measured using clinical thermometers or laboratory thermometers, respectively. They are not intended to keep track of the daily minimum and maximum temperatures.

24. Correct option: (D) Tropical rainforest

A tiger is a carnivorous animal that fits the definition; it is distinguished by its striking stripes and extraordinary speed and agility when pursuing prey. In Asia, grasslands and tropical rainforests are the main habitats of tigers. They are strong predators that have evolved to live in these luxuriant, thick forest habitats. These animals have stripes on their body that help them to camouflage in the dense vegetation of the rainforest. They also have sharp claws and teeth that help them to catch their prey.
The Arctic, which is significantly dissimilar to the tropical jungle in terms of habitat, is where polar bears may be found. Tropical rainforests are often very hot and humid, while deserts are typically very hot and dry. Ocean-dwelling animals have gills or a thick covering of blubber as adaptations to help them survive in the saline water.

25. Correct option: (C) Long grasping tail

The qualities of all the other options (A, B, and D) are often associated with animals that have evolved to survive in frigid climates, such the Arctic or polar regions, which makes C) the odd one out.
Polar bears and seals employ this adaptation, known as blubber, to help them insulate themselves. A) Thick layer of fat under the skin.
B) White fur: Several animals have developed white fur as a form of camouflage to help them blend in with their surroundings.
D) This adaptation is typically seen in animals with large, broad feet and sharp claws, such as polar bears, which makes it simpler for them to walk about, grab icy surfaces, and effectively catch prey.

Animals living in cold climates are not often thought to have long, gripping tails. For animals that dwell in trees or deep woods and depend on holding on to branches for support or movement, a long gripping tail may be more appropriate. It is not a trait frequently observed in creatures that have evolved to cold regions.

26. Correct option: (C) Mix with the surroundings

Tigers and leopards have evolved to have stripes on their body, which gives them the ability to blend in with their environment and serve as an excellent kind of camouflage. Their ability to blend in and approach their target covertly is especially helpful for their hunting tactics.

In the dappled light and shadows of their native environments, such as grasslands and woodlands, the stripes blur the shape of their bodies, making it more difficult for their prey to identify them. Their ability to blend in with their surroundings and approach their victim undetected until they are prepared to strike is crucial to their effectiveness as predators.

27. Correct option: (B)

28. Correct option: (D) a – iv; b – ii; c – i; d – iii

a. Red-eyed frog has iv. Sticky pads on feet. They can climb trees because to this adaptability. Arboreal creatures, red-eyed frogs spend the most of their time in trees. Their feet include sticky pads that allow them to effortlessly move around in trees and hold onto branches.

b. Penguin has ii. Streamline body. This modification enhances their capacity for swift and effective swimming. Penguins can hunt fish and other prey using their streamlined bodies since they are skilled divers and swimmers.

c. Tiger has i. Very sensitive hearing. They have the capacity to hear noises that are imperceptible to humans. This modification makes it easier for them to discover prey and recognize predators.

d. Lion-tailed macaque has iii. Silver-white mane. They can attract mates thanks to this modification. Additionally, the main makes lions appear bigger and more frightening, which can aid in fending off predators.

So, the answer is D. a – iv; b – ii; c – i; d – iii.

29. Correct option: (A) a – iv; b – iii; c – ii; d – i

a) Elements of weather - iv) Temperature, humidity: The term "elements of weather" refers to a variety of elements, such as temperature and humidity, that make up the atmospheric conditions.

b) Cause of all changes - iii) Sun: The sun is the main influence on Earth's weather and climate. All alterations in weather patterns and atmospheric conditions are influenced by its energy.

c) Tropical region - ii) Hot and wet: High temperatures and high humidity combine to create a hot and humid environment in tropical locations.

d) Desert climate - i) Hot and dry: Low humidity and high temperatures create a dry and warm atmosphere in desert settings.

30. Correct option: (B) (i), (iii)

The conditions from the above list that are responsible for the presence of a large number of plants and animals in tropical rainforests are:

(i) Humid and hot climate: Tropical rainforest’s humid and hot climates are ideal for the development and spread of a wide range of plant species. High temperatures and humidity encourage quick plant development, which results in a varied and thick forest canopy.

(iii) Abundant rainfall: Tropical rainforests get an abundance of rainfall, which guarantees a steady supply of water, which is essential for the survival and development of both plants and animals. It supports a wide range of plant growth and offers home to a large number of animal species.

9. Garbage in and Garbage Out
Solutions

LEVEL – I

1. Correct Option (B)

The right response is (b) Eggshells, tea leaves, and fruit and vegetable peels. Decomposing organic materials into nutrient-rich humus through the process of composting can enhance the soil's quality. It's crucial to use organic material that will decompose naturally in this process. Eggshells, tea leaves, and fruit and vegetable peels are examples of items high in organic matter and nitrogen that are advantageous for composting. They give the compost pile's microorganisms the nutrients they need to thrive and convert organic matter into compost that is useful. In contrast, materials like plastic, polythene bags, paper, tin foil, and leather are included in options (a), (c), and (d), which are not suitable for composting because they are not biodegradable or because they can introduce dangerous contaminants into the compost.

2. Correct Option (B)

Due to their reduced risk of flooding, landfills are typically found in low-lying areas. A landfill's waste may become contaminated by flooding, and the surrounding area may become contaminated as well. Furthermore, low-lying areas are frequently less desirable for other purposes, like agriculture or habitation.

In order to keep waste contained and keep it from polluting the environment, landfills were created. To stop groundwater from leaking into the waste, they have plastic or clay linings. In order to keep the waste from blowing away or luring pests, it is also covered with soil.

Although landfills are a necessary component of waste disposal, they can have detrimental effects on the environment. They have the ability to release methane, a greenhouse gas, into the air. They can also contaminate groundwater and soil. However, there are ways to minimize the environmental impacts of landfills, such as using methane capture and treatment systems.

3. Correct Option (C)

The answer is (c) Eggshells, vegetable and fruit peels, and tea leaves. These materials are great options for composting because they are organic, can decompose naturally, and can add vital nutrients to the compost. Tea leaves add nitrogen to the mix, fruit and vegetable peels offer a variety of organic matter, and eggshells contribute calcium. The items listed in option (c) are perfect components for creating nutrient-rich compost that can be used to improve the quality of the soil for gardening and farming, while options (a), (b), and (d) include non-biodegradable or unsuitable materials for composting. Composting is an eco-friendly way to recycle kitchen and garden waste.

4. Correct Option (D)

The items in the options list all contribute to waste in various ways. Despite having a medical use, the syringe a doctor uses turns into hazardous waste and needs to be properly disposed of. A soft drink can and a food wrapper are examples of single-use items that, after being used, typically end up as litter or non-biodegradable waste in landfills. In order to reduce their negative effects on the environment, all three of the items mentioned in the options should be responsibly disposed of or recycled whenever possible.

5. Correct Option (C)

Recycling is a waste management technique that involves processing and using materials from discarded items to make new products, not as a means of disposing of trash. Recycling is the process of gathering, sorting, and reusing waste materials like paper, glass, plastic,

and metal to create new products with less raw material consumption and less environmental impact.

The other three choices—landfill, vermicomposting, and composting—all involve getting rid of waste. In a landfill, waste is disposed of in a specific location. Worms are used in the vermicomposting process, which breaks down organic waste. Utilizing microorganisms to break down organic waste is known as composting.

Recycling is a more environmentally friendly method of waste disposal than composting, vermicomposting, or landfilling. Recycling promotes conservation.

6. Correct Option (D)

Garbage goes through several stages of decay when it rots or decomposes, during which time organic material is broken down into simpler compounds. When the waste has fully decomposed and no longer emits any detectable odors, the process is deemed to be finished. The organic materials at this point have developed into a stable, humus-rich compost or soil conditioner that can be used to improve the soil for farming or gardening. When all the organic material has decomposed and any offensive odors have disappeared, leaving behind a useful and odorless resource for soil improvement, properly managed compost piles reach this stage.

7. Correct Option (C)

Green is typically the color of the trash can used for biodegradable or organic waste, such as food scraps, fruit and vegetable peels, and yard waste. The proper segregation of waste is facilitated by the widespread use of this color-coding system for trash cans throughout the world. Biodegradable waste is placed in green bins because it can be composted to produce valuable organic matter for soil improvement. Utilizing different colored bins makes it simpler for people to correctly sort their trash, encouraging recycling and eco-friendly waste management techniques.

8. Correct Option (A)

Sewage is the term for the domestically produced liquid waste. Wastewater from different domestic sources, such as toilets, sinks, showers, and washing machines, is referred to as sewage. Human waste, soaps, detergents, and other contaminants are all mixed together in it. Before sewage is released into the environment or recycled as treated water for various uses, it must be properly collected and treated to remove contaminants and pathogens. Since untreated sewage can present serious environmental and health risks, effective sewage treatment is essential to prevent water pollution and safeguard public health.

9. Correct Option (C)

Wastepaper is the option (c) on the list that can be recycled. Paper waste, including office paper, cardboard, and magazines, can be gathered, processed, and recycled to create new paper goods. Paper recycling lessens the environmental impact of paper production by saving energy and trees. On the other hand, organic materials like animal waste and kitchen scraps can be composted rather than recycled through conventional recycling processes, while leather shoes (a) are typically not recyclable due to their complex materials and construction.

10. Correct Option (A)

One type of waste that can be recycled is plastic. Plastic is a non-biodegradable substance that takes hundreds of years to break down naturally, in contrast to organic wastes like onion peel, eggshell, and banana peel. Recycling plastic saves energy and resources that would otherwise be used in the manufacture of new plastic, as well as reducing the amount of waste that is dumped in landfills or the ocean. Recycling plastic waste allows it to be turned into fresh goods like bottles, containers, and even clothing, which promotes a sustainable and circular economy.

On the other hand, onion peels (b), eggshells (c), and banana peels (d) are organic waste materials that are typically composted rather than recycled.

11. Correct Option (D)

All of these things should be disposed of according to standard waste disposal procedures in the garbage. Because it is frequently made of difficult-to-separate and recyclable mixed materials, the wrapper of a biscuit pack (a) is typically not recyclable. The tin of a cold drink (b) should also be thrown out in the trash because recycling it might be challenging given that it has an interior lining and might still contain beverage remnants. In order to avoid contamination and potential harm, the syringe a doctor used (c) should be properly disposed of as medical waste by adhering to safety protocols. Therefore, for safe and responsible disposal, each of these items should be placed in the proper waste containers.

12. Correct Option (A)

Earthworms play a crucial role in the composting process and the production of nutrient-rich organic matter, or manure. They consume organic waste products like food scraps and decaying plant matter, breaking them down into castings that are rich in nutrients and an important part of natural fertilizers and manure. In contrast to earthworms, which are typically involved in the decomposition or composting of organic matter, houseflies, eagles, and crows are not. Rather, they fill various ecological niches and play different roles in the environment, such as scavenging or hunting for food.

13. Correct Option (C)

Making compost through vermicomposting is option (c). With the help of vermicomposting, it is possible to efficiently and sustainably dispose of leaves that fall from trees by converting them into nutrient-rich compost. Earthworms are used in vermicomposting to convert organic matter, such as leaves, into castings that are rich in nutrients. This process reduces waste while also producing a valuable soil conditioner that can improve plant growth and soil fertility. Burning leaves is not environmentally friendly and can release harmful pollutants into the air, whereas boiling and drying leaves to make compost is an energy-intensive process that is less effective than vermicomposting. Vermicomposting is thus the most advantageous and environmentally responsible method of getting rid of tree debris.

14. Correct Option (A)

Usually, waste from cities is collected and disposed of at designated landfill sites. For the safe and controlled disposal of solid waste produced by urban areas, landfills are engineered buildings. These locations are carefully managed to reduce their negative effects on the environment and prevent soil and groundwater contamination. To manage and contain the waste, it is deposited, compacted, and covered with layers of soil or other materials in these landfill areas, where it is gathered from homes, businesses, and other sources. The other alternatives, near every colony (d), inside ponds and lakes (c), and along riversides (b), are inappropriate and environmentally harmful places to dispose of municipal waste because they can harm ecosystems and cause pollution.

15. Correct Option (A)

Waste paper is a prime candidate for recycling because it can be recycled using a variety of techniques to make new paper products. Recycling used paper reduces the need for new wood pulp, protecting forests and other natural resources while reducing energy use during the papermaking process. Conversely, leather shoes (b) may be donated or used for another purpose but are typically not recycled. Since they naturally decompose into nutrient-rich compost that can be used to improve soil for gardening or agriculture, animal waste (c) and kitchen waste (d) are organic materials that can be composted rather than recycled.

16. Correct Option (C)

Due to its composition, which prevents it from decomposing naturally over time, aluminium foil is a waste that cannot decompose. Aluminum foil is made from a metal and does not decompose in the same way as paper, which is made from plant fibres and has that ability. Therefore, aluminium foil that is thrown away in the environment will do long-term environmental harm and contribute to pollution. Aluminium foil is the non-biodegradable waste, so far, among the options presented.

In contrast, options like paper (a), faecal matter (b), and cotton (d) are biodegradable materials. Paper and cotton are derived from plant fibers and can decompose over time when exposed to environmental conditions, while faecal matter is organic waste that can also biodegrade and return nutrients to the soil.

17. Correct Option (A)

The acronym "3'R" stands for three critical environmental preservation and sustainable waste management principles. By using fewer resources and products, "reduce" encourages waste minimization, "reuse" encourages the practice of reusing items multiple times to increase their lifespan, and "recycle" emphasizes the process of converting materials and products into new items to lessen the need for fresh raw materials and energy. Together, these three ideas support a more environmentally responsible and sustainable method of resource conservation, waste management, and environmental protection.

18. Correct Option (D)

Due to the fact that bread is an organic substance that can naturally decompose, it will not pollute the soil. In the long run, bread left in the soil will decompose and supply nutrients to the nearby plants and animals. Thermocol, plastic, and aluminium foil, on the other hand, are all non-biodegradable substances that take a long time to break down. These substances have a long half-life in the soil and can harm the environment and cause pollution. Therefore, the only option among those presented that won't contaminate the soil is bread.

19. Correct Option (A)

Microorganisms like bacteria and fungi are primarily responsible for rotting, or decomposition. The breakdown of organic matter, such as dead plants, animals, and other organic materials, is greatly aided by these tiny organisms. Enzymes that are secreted by microbe’s aid in the breakdown of complex organic compounds into less complex ones so that the microbes can then absorb the simpler compounds as nutrients. This decomposition procedure is a crucial component of nature's waste disposal system and is necessary for recycling nutrients back into the ecosystem. Although insects (b) and ants (c) might make a small contribution to the deterioration of organic matter, microbes actually play the majority of the role in rotting. Amoeba (d) is a single-celled organism that does not typically participate in the rotting or decomposition process.

20. Correct Option (B)

Typically, (b) Blue bins are used to collect recyclable materials. In numerous recycling programs all over the world, blue bins are frequently used for the collection of recyclable materials. Paper, cardboard, plastic bottles, glass containers, and metal cans are all acceptable materials to place in these bins because they can all be recycled to make new products. By making it simple for people to distinguish between recyclables and non-recyclable waste, the color coding encourages recycling and ethical waste disposal practices. Regional differences in bin color may exist, but recycling bins are typically blue in color.

21. Correct Matching: a-iv, b-v, c-ii, d-iii, e-i

(a) Landfill - (iv) Land where garbage is thrown: Landfills are places set aside for disposing of solid waste, including garbage. They are substantial locations where waste is dumped and frequently covered with dirt or other materials.

(b) Vermicomposting - (v) Method of preparing compost: Vermicomposting is a composting technique in which organic matter, such as kitchen scraps, is broken down by worms, usually red worms, to create nutrient-rich compost.

(c) Red worms - (ii) Form compost: Earthworms of the red variety are used in vermicomposting. They consume organic waste and aid in the breakdown of materials to create compost.

(d) Recycling - (iii) Making things from waste: Recycling is the process of gathering and processing waste materials in order to transform them into new products. Waste materials are converted into manufacturing raw materials in this process.

(e) Plastic - (i) Give out harmful gases on burning: Burning plastic may release toxic gases into the atmosphere that could harm human health and contribute to air pollution. Due to the risks to both the environment and human health, burning plastic is generally discouraged.

LEVEL – II

1. Correct Option (C)

The answer is (c) Waste material that is disposed by humans. Organic waste that can naturally decompose over time with the aid of microorganisms like bacteria and fungi is classified as biodegradable waste. Food scraps, vegetable peels, fruit rinds, and other organic matter are frequently found in kitchen waste. These materials can be turned into nutrient-rich compost, which can be used as a natural fertilizer to improve soil, when properly disposed of, such as through composting. As a result, kitchen waste can be used for recycling and sustainable waste management strategies, which helps to lessen the amount of organic waste that is dumped in landfills and promotes the health of the soil and plant growth.

2. Correct Option (A)

Kitchen waste is considered as "biodegradable" waste. Organic waste that can naturally decompose over time with the aid of microorganisms like bacteria and fungi is classified as biodegradable waste. Food scraps, vegetable peels, fruit rinds, and other organic matter are frequently found in kitchen waste. These materials can be turned into nutrient-rich compost, which can be used as a natural fertilizer to improve soil, when properly disposed of, such as through composting. As a result, kitchen waste can be used for recycling and sustainable waste management strategies, which helps to lessen the amount of organic waste that is dumped in landfills and promotes the health of the soil and plant growth.

3. Correct Option (B)

Typically, garbage is gathered by cleaning crews and transported to landfill sites, which are low-lying areas set aside for the disposal of solid waste. Landfills are constructed buildings intended for the safe management and containment of waste. To reduce the environmental impact and control odors, waste is compacted and covered with soil or other materials before being dumped in designated areas by garbage trucks at landfills. To avoid pollution, safeguard groundwater, and guarantee that waste is appropriately contained and handled in an environmentally responsible way, landfills are carefully managed. Septic tanks and sewage tanks, on the other hand, are intended for the treatment of wastewater rather than the disposal of solid waste.

4. Correct Option (C)

A landfill can be turned into a park once it is entirely full and no longer useful for the disposal of waste. This procedure, which is a component of landfill reclamation, is regarded as an eco-friendly way to use land that had previously been used for waste disposal. A properly closed and covered landfill can be converted into a park, a green space, or other public amenity, serving as a valuable resource for the neighborhood. The procedure typically entails landscaping the area to make it safe and aesthetically pleasing for public use, covering the landfill with soil, and installing a protective liner. Former landfills being turned into parks not only helps to improve the local environment but also encourages sustainable land use and neighborhood growth.

5. Correct Option (A)

Dry leaves fall under the "biodegradable" waste category. Organic materials that naturally decompose over time with the aid of microorganisms like bacteria and fungi are referred to as biodegradable waste. Dry leaves can be composted or allowed to naturally decompose in the environment because they are organic plant material. They break down into nutrient-rich humus, which can be applied to the soil as natural fertilizer, when they decompose. As a result, dry leaves become a valuable resource for composting and a sustainable method of recycling organic material, lowering the amount of waste that is dumped in landfills.

Waste that cannot be broken down by microorganisms is referred to as non-biodegradable waste. Plastic, glass, and metal waste are a few non-biodegradable waste examples.

Composed waste is waste that has undergone microbial degradation. Compost is a nutrient-rich soil enhancement material that can be used to enhance plant health. Waste that can be divided into different categories, such as recyclables, compostables, and non-recyclables, is referred to as gradeable waste.

6. Correct Option (B)

A plastic bag is classified as a "non-biodegradable" waste. Materials that do not naturally decompose or break down into harmless substances over time when disposed of in the environment are considered non-biodegradable. Synthetic polymers with high resistance to microbial decomposition, like polyethylene, are the main constituents of plastic bags. Because of this, when plastic bags are thrown away, they can stay in the environment for a long time and cause litter, pollution, and harm to wildlife. To lessen their damaging effects on the environment, there is an increasing focus on reducing the use of plastic bags and promoting their recycling or proper disposal.

7. Correct Option (B)

The rotting and conversion of some material into manure is called "composting." Composting is a natural process in which microorganisms like bacteria, fungi, and enzymes transform organic waste products like kitchen scraps, yard waste, and other biodegradable materials into nutrient-rich humus-like material. The conditions for decomposition are created by layering, turning, and aerating the organic materials in a controlled environment, usually a compost pile or bin. These materials eventually break down into compost, a beneficial fertilizer and soil conditioner that can be used to strengthen the soil and promote plant growth. Therefore, composting is an environmentally friendly and sustainable method for turning organic waste into nutrient-rich manure, and it aids in lowering the amount of waste that ends up in landfills while fostering soil sustainability.

8. Correct Option (A)

The organisms in vermicompost, specifically a species known as Eisenia fetida or red wiggler worms, are essential for converting organic waste into usable manure. Due to their voracious appetite for organic matter, including food scraps and other decomposing plant materials, these worms are ideal for composting. As the redworms eat this organic waste,

their digestive systems break it down, and their castings—their excrement—become rich in nutrients and advantageous microorganisms. These "black gold" worm castings are excellent organic fertilizers and soil conditioners, adding vital nutrients and enhancing soil structure for more robust plant growth. Redworms are therefore crucial to the vermicomposting process, assisting in the transformation of organic

9. Correct Option (D)

Plastics, metals, and glass fragments are among the recyclables that are frequently collected in the blue bin. This container is a part of a recycling programme that separates and recycles these materials in an effort to cut waste and save resources. Glass pieces refer to glass bottles and containers, while plastics cover a wide range of plastic goods and packaging, while metals include things like steel containers and Aluminium cans. By putting all of these recyclable materials in the blue bin, people can help the recycling process, in which waste is gathered, sorted, and processed to create new products, thereby promoting resource conservation and environmental sustainability. Option (d), "All the given options," is the right response because it includes all of the collected recyclable materials.

10. Correct Option (B)

Dry leaves are an extremely biodegradable form of organic plant material. Dry leaves easily decompose when added to soil thanks to the action of microorganisms like bacteria and fungi. The soil is enriched for plant growth by these microorganisms as they convert the leaves into organic matter and release nutrients. In contrast, non-biodegradable alternatives like plastic, glass fragments, and nylon strings are difficult for the soil to break down. They have a long shelf life and can pollute the environment and harm ecosystems. Dry leaves are an excellent resource for composting and soil enrichment because they can easily and naturally decompose in the soil as waste.

11. Correct Option (B)

Red worms, also referred to as earthworms, lack teeth, unlike people. Instead, they can grind and digest their food with the aid of a special organ called a gizzard. The gizzard is a muscular organ in the worm's digestive tract that is a part of its digestive system. The gizzard contracts after ingesting small mineral particles and grit, which it uses to grind the food when the worms consume organic matter like leaves or other decomposing plant material. This physical breaking down of the food into smaller pieces facilitates the further breakdown and digestion of the organic matter by the worms' digestive enzymes. So, the gizzard plays a crucial role in the earthworm's digestion process by aiding in the mechanical breakdown of food.

Animals use their teeth to tear and chew their food. Red worms are unable to tear or chew their food because they lack teeth. Additionally, they lack the radula, an organ resembling a tongue that some animals use to scrape food into their mouths. Red worms, on the other hand, eat their food whole and rely on their gizzard to grind it.

12. Correct Option (C)

The correct answer is (c) Reuse. Reusing something means using it without significantly altering it or processing it extensively. Finding new applications for an item after its initial use—either in its current form or with only minor modifications—instead of throwing it away after use. Making the most of already-existing materials promotes resource conservation, lowers waste generation, and increases product lifespan. In contrast, choices like "Reduce" put the emphasis on cutting back on consumption, "Recycle" entails using old materials to make new products, and "Redraw" has nothing to do with using or modifying existing things. Reusing something means using it without significantly processing it or changing it, which is consistent with the definition of reuse.

13. Correct Option (B)

Recycling is the process of using waste materials that have been used up or discarded to make new products or materials. To do this, recyclable materials like paper, glass, plastics, and metals must be gathered, sorted, cleaned, and converted into raw materials that can be used to create new goods. Recycling is a crucial component of sustainable waste management because it aids in resource conservation, lowers energy usage, and lessens the environmental impact of producing new materials. Therefore, "Recycle" is the term that specifically refers to processing something to create a new product from previously used or discarded materials among the options offered.

14. Correct Option (D)

Plastic should not be thrown down drains, into water tanks, or out of water outlets as it may obstruct the flow. Plastic waste that enters drains has a tendency to build up and clog the drainage system, which can cause waterlogging and flooding during periods of heavy rain. Plastic waste can contaminate the water supply and make it more challenging to filter and purify the water if it is put into water tanks. Similar to plastic waste, obstructions in water outlets can stop water flow, causing backups and possibly damaging infrastructure. Because of this, disposing of plastic waste in these places is not only bad for the environment but can also interfere with key services and result in a number of issues.

15. Correct Option (A)

When compared to making paper from scratch, recycling paper is an environmentally friendly strategy that not only lessens the need to clear forests of trees but also saves energy and resources. When paper is recycled, it goes through a process that makes new paper products out of it, extending its useful life and reducing the amount of waste that is dumped in landfills. While recycling is particularly effective in ensuring that paper materials are repurposed and made into new items, contributing to a more sustainable and environmentally friendly approach to managing paper waste, reusing and composting paper are also beneficial methods for reducing waste.

16. Correct Option (C)

The organic waste that can naturally decompose over time is called biodegradable waste, and it is kept in the green dust bin. This includes things like leftover food, garbage from the kitchen, grass clippings from the garden, and other organic materials that decompose into compost or can be recycled naturally. In order to encourage environmentally friendly disposal techniques, like composting or recycling, rather than sending biodegradable waste to landfills where it contributes to the production of methane gas and other environmental issues, green bins are typically designated for this type of waste. As a result, option (c), "biodegradable waste," is the appropriate response because it fits with the objective of green dust bins.

17. Correct Option (C)

Burning waste causes air pollution, as indicated by option (c). Burning waste materials releases dangerous pollutants into the atmosphere, especially if they contain plastics, paper, or other synthetic materials. Particulate matter, carbon monoxide, Sulphur dioxide, nitrogen oxides, volatile organic compounds, and dangerous chemicals are just a few examples of these pollutants. Humans who breathe in these pollutants may experience serious respiratory and cardiovascular problems as well as other negative health effects. The fact that burning waste releases greenhouse gases also emphasizes the harm that this type of pollution does to the environment by causing climate change. As a result, burning waste is a major cause of air pollution.

18. Correct Option (B)

We do not use plastic bags to store eatables, as indicated by option (b). Because plastic bags are made of synthetic materials that can leach dangerous chemicals into food, they are not appropriate for storing food items. When these chemicals come into contact with edible products, such as phthalates and bisphenol A (BPA), health risks may result. As opposed to plastic bags, which are more likely to contaminate food than jute, nylon, and polyester bags, these materials are generally thought to be safer options for storing food. Therefore, due to the potential health risks associated with their use, plastic bags are not advised for food storage.

19. Correct Option (D)

All the options listed will add to garbage. Kitchen waste, such as food scraps and leftovers, contributes to organic waste in garbage. The wrapper of a biscuit is typically made of non-biodegradable materials like plastic or Aluminium foil, which also ends up in the garbage. Similarly, the tin of a cool drink, made of metal or Aluminium, is not biodegradable and becomes part of the garbage when discarded. Therefore, each of these options, (a) Kitchen waste, (b) Wrapper of a biscuit, and (c) Tin of a cool drink, adds to the overall volume of garbage when they are disposed of improperly.

20. Correct Option (C)

Wastepaper is a type of paper that has been used previously but can still be processed to create new paper products. Recycling used paper helps preserve forests and uses less energy and resources than manufacturing new paper from scratch. In contrast, animal and kitchen waste can be composted to produce soil rich in nutrients, but these organic materials are not typically recycled in the same way as paper products. Wastepaper is the only option from the ones provided that can be recycled to make new products.

21. Correct Option (C)

The term "3'R" is a sustainability concept that encourages responsible resource management and environmental conservation.

Reduce: This entails cutting back on resource consumption and waste production. It encourages people and companies to use resources more wisely, which can include buying energy-efficient appliances, saving water, and using fewer single-use items.

Reuse: Reuse encourages reusing goods or resources rather than discarding them after a single use. This can entail doing things like donating clothing and goods rather than throwing them away and using reusable containers and shopping bags.

Recycle: Recycling is the process of gathering and processing materials into new products that would otherwise be thrown away as waste. It helps keep resources in check, uses less energy, and keeps waste out of landfills.

22. Correct Option (B)

Microorganisms like bacteria, fungi, and other microbes play a major role in the process of rotting, also known as decomposition. These tiny creatures are essential for the transformation of organic matter, such as excreta from dead animals and plants, into simpler substances. They accomplish this by secreting enzymes that disassemble intricate organic molecules into simpler parts that the ecosystem can absorb and recycle. Microbes are in charge of the environment's natural recycling of nutrients and organic materials, which makes it possible for crucial elements like carbon and nitrogen to return to the soil and support plant growth and ecological balance. Therefore, decay is a biological process fueled by microbes.

23. Correct Option (C)

Organic materials like dry leaves naturally decompose, supplying the soil with vital nutrients as they do so. Dry leaves are preferable to drainage water (option b), which can

introduce pollutants and contaminants into the soil, and plastic (option a), which is non-biodegradable and can persist in the soil for hundreds to thousands of years. In addition to decomposing, they also improve soil structure, moisture retention, and promote advantageous microbial activity. Due to their advantageous qualities, dry leaves are actually widely used by farmers and gardeners as a form of mulch or as a component of compost. Dry leaves therefore benefit the environment rather than polluting the soil.

24. Correct Option (C)

The correct answer is (c) Nutrients. Compost that is rich in nutrients is created when organic waste, such as plant waste and food scraps, completely rots during the composting process. Nitrogen, phosphorus, potassium, and a variety of micronutrients are just a few of the many nutrients that compost is a great source of. As the compost decomposes more, these nutrients are gradually released, giving plants a natural and balanced supply of nutrition. Compost adds these crucial nutrients to the soil, enhancing its structure and encouraging the growth of healthy plants and higher yields. Option (c) is the best option because composting allows garbage to completely rot while also giving plants valuable nutrients.

25. Correct Option (B)

When added to soil, compost is a valuable organic material that enriches it with organic matter and crucial nutrients. Compost is well known for its capacity to enhance soil structure, boost nutrient availability, and encourage advantageous microbial activity. These characteristics make the soil more fertile, which makes it more conducive to plant growth. Increased crop yields and healthier gardens result from the conditions and nutrients that fertile soil offers plants for healthy development. So, improving soil's fertility and overall productivity can be accomplished by adding compost to the soil.